12

Static Equilibrium and Elasticity

CHAPTER OUTLINE

12.1 Analysis Model: Rigid Object in Equilibrium

12.2 More on the Center of Gravity

12.3 Examples of Rigid Objects in Static Equilibrium

12.4 Elastic Properties of Solids

* An asterisk indicates an item new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ12.1 Answer (b). The skyscraper is about 300 m tall. The gravitational field (acceleration) is weaker at the top by about 900 parts in ten million, by on the order of 10^{-4} times. The top half of the uniform building is lighter than the bottom half by about $(1/2)(10^{-4})$ times. Relative to the center of mass at the geometric center, this effect moves the center of gravity down, by about $(1/2)(10^{-4})(150\text{ m}) \sim 10$ mm.

OQ12.2 Answer (c). Net torque = $(50\text{ N})(2\text{ m}) - (200\text{ N})(5\text{ m}) - (300\text{ N})x = 0$; therefore, $x = 3$ m.

OQ12.3 Answer (a). Our theory of rotational motion does not contradict our previous theory of translational motion. The center of mass of the object moves as if the object were a particle, with all of the forces applied there. This is true whether the object is starting to rotate or not.

OQ12.4 Answer (d). In order for an object to be in equilibrium, it must be in both translational equilibrium and rotational equilibrium. Thus, it must meet two conditions of equilibrium, namely $\vec{\mathbf{F}}_{\text{net}} = 0$ and $\vec{\tau}_{\text{net}} = 0$.

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OQ12.5 Answer (b). The lower the center of gravity, the more stable the can. In cases (a) and (c) the center of gravity is above the base by one-half the height of the can. In case (b), the center of gravity is above the base by only a bit more than one-quarter of the height of the can.

OQ12.6 Answer (d). Using the left end of the plank as a pivot and requiring that $\sum \tau = 0$ gives

$$-mg(2.00\text{ m}) + F_2(3.00\text{ m}) = 0$$

or

$$F_2 = \frac{2mg}{3} = \frac{2(20.0\text{ kg})(9.80\text{ m/s}^2)}{3} = 131\text{ N}$$

OQ12.7 Answer: $\tau_D > \tau_C > \tau_E > \tau_B > \tau_A$. The force exerts a counterclockwise torque about pivot *D*. The line of action of the force passes through *C*, so the torque about this axis is zero. In order of increasing negative (clockwise) values come the torques about *F*, *E* and *B* essentially together, and *A*.

OQ12.8 Answer (e). In the problems we study, the forces applied to the object lie in a plane, and the axis we choose is a line perpendicular to this plane, so it appears as a point on the force diagram. It can be chosen anywhere. The algebra of solving for unknown forces is generally easier if we choose the axis where some unknown forces are acting.

OQ12.9 (i) Answer (b). The extension is directly proportional to the original dimension, according to $F/A = Y\Delta L/L_i$.

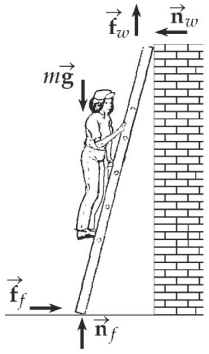
(ii) Answer (e). Doubling the diameter quadruples the area to make the extension four times smaller.

OQ12.10 Answer (b). Visualize the ax as like a balanced playground seesaw with one large-mass person on one side, close to the fulcrum, and a small-mass person far from the fulcrum on the other side. Different masses are on the two sides of the center of mass. The mean position of mass is not the median position.

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ANSWERS TO CONCEPTUAL QUESTIONS

CQ12.1 The free-body diagram demonstrates that it is necessary to have friction on the ground to counterbalance the normal force of the wall and to keep the base of the ladder from sliding. If there is friction on the floor *and* on the wall, it is not possible to determine whether the ladder will slip, from the equilibrium conditions alone.



ANS. FIG. CQ12.1

CQ12.2 A V-shaped boomerang, a barstool, an empty coffee cup, a satellite dish, and a curving plastic slide at the edge of a swimming pool each have a center of mass that is not within the bulk of the object.

CQ12.3 (a) Consider pushing up with one hand on one side of a steering wheel and pulling down equally hard with the other hand on the other side. A pair of equal-magnitude oppositely-directed forces applied at different points is called a couple.

(b) An object in free fall has a nonzero net force acting on it, but a net torque of zero about its center of mass.

CQ12.4 When one is away from a wall and leans over, one's back moves backward so the body's center of gravity stays over the feet. When standing against a wall and leaning over, the wall prevents the backside from moving backward, so the center of gravity shifts forward. Once your CG is no longer over your feet, gravity contributes to a nonzero net torque on your body and you begin to rotate.

CQ12.5 If an object is suspended from some point and allowed to freely rotate, the object's weight will cause a torque about that point unless the line of action of its weight passes through the point of support. Suspend the plywood from the nail, and hang the plumb bob from the nail. Trace on the plywood along the string of the plumb bob. The plywood's center of gravity is somewhere along that line. Now suspend the plywood with the nail through a different point on the plywood, not along the first line you drew. Again hang the plumb bob from the nail and trace along the string. The center of gravity is located halfway through the thickness of the plywood under the intersection of the two lines you drew.

CQ12.6 She can be correct. Consider the case of a bridge supported at both ends: the sum of the forces on the ends equals the total weight of the bridge. If the dog stands on a relatively thick scale, the dog's legs on

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the ground might support more of its weight than its legs on the scale. She can check for and if necessary correct for this error by having the dog stand like a bridge with two legs on the scale and two on a book of equal thickness—a physics textbook is a good choice.

CQ12.7 Yes, it can. Consider an object on a spring oscillating back and forth. In the center of the motion both the sum of the torques and the sum of the forces acting on the object are (separately) zero. Again, a meteoroid flying freely through interstellar space feels essentially no forces and keeps moving with constant velocity.

CQ12.8 Shear deformation. Its deformations are parallel to its surface.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

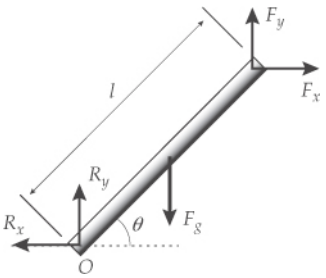
Section 12.1 Analysis Model: Rigid Object in Equilibrium

P12.1 Use distances, angles, and forces as shown in ANS. FIG. P12.1. The conditions of equilibrium are:

$$\sum F_y = 0 \Rightarrow \boxed{F_y + R_y - F_g = 0}$$

$$\sum F_x = 0 \Rightarrow \boxed{F_x - R_x = 0}$$

$$\sum \tau = 0 \Rightarrow \boxed{F_y \ell \cos\theta - F_g\left(\frac{\ell}{2}\right)\cos\theta - F_x \ell \sin\theta = 0}$$



ANS. FIG. P12.1

P12.2 Take torques about P, as shown in ANS. FIG. P12.2.

$$\sum \tau_p = -n_O\left[\frac{\ell}{2} + d\right] + m_1 g\left[\frac{\ell}{2} + d\right] + m_b g d - m_2 g x = 0$$

We want to find x for which $n_O = 0$:

$$x = \frac{\left(m_1 g + m_b g\right)d + m_1 g\dfrac{\ell}{2}}{m_2 g} = \frac{\left(m_1 + m_b\right)d + m_1\dfrac{\ell}{2}}{m_2}$$

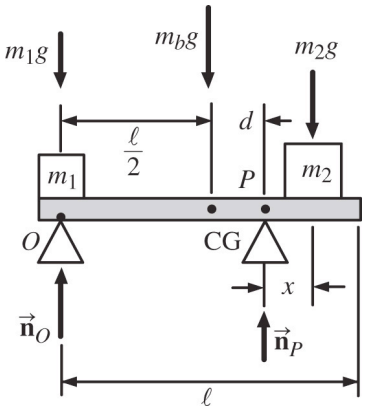
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For the values given:

$$x = \frac{(m_1 + m_b)d + m_1\dfrac{\ell}{2}}{m_2}$$

$$x = \frac{(5.00\text{ kg} + 3.00\text{ kg})(0.300\text{ m}) + (5.00\text{ kg})\dfrac{1.00\text{ m}}{2}}{15.0\text{ m}}$$

$$x = 0.327\text{ m}$$



ANS. FIG. P12.2

The situation is impossible because x is larger than the remaining portion of the beam, which is 0.200 m long.

Section 12.2 More on the Center of Gravity

P12.3 The coordinates of the center of gravity of piece 1 are

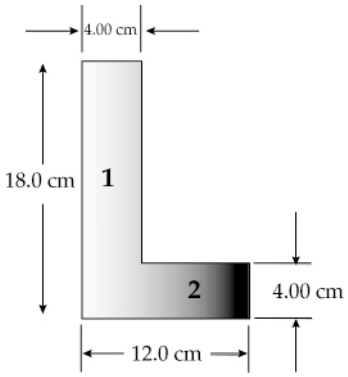
$$x_1 = 2.00\text{ cm} \text{ and } y_1 = 9.00\text{ cm}$$

The coordinates for piece 2 are

$$x_2 = 8.00\text{ cm} \text{ and } y_2 = 2.00\text{ cm}$$

The area of each piece is

$$A_1 = 72.0\text{ cm}^2 \text{ and } A_2 = 32.0\text{ cm}^2$$



ANS. FIG. P12.3

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And the mass of each piece is proportional to the area. Thus,

$$x_{CG} = \frac{\sum m_i x_i}{\sum m_i} = \frac{(72.0\ \text{cm}^2)(2.00\ \text{cm}) + (32.0\ \text{cm}^2)(8.00\ \text{cm})}{72.0\ \text{cm}^2 + 32.0\ \text{cm}^2}$$

$$= \boxed{3.85\ \text{cm}}$$

and

$$y_{CG} = \frac{\sum m_i y_i}{\sum m_i} = \frac{(72.0\ \text{cm}^2)(9.00\ \text{cm}) + (32.0\ \text{cm}^2)(2.00\ \text{cm})}{104\ \text{cm}^2}$$

$$= \boxed{6.85\ \text{cm}}$$

P12.4 The definition of the center of gravity as the average position of mass in the set of objects will result in equations about x and y coordinates that we can rearrange and solve to find where the last mass must be.

From $\vec{\mathbf{r}}_{CG} = \dfrac{\sum m_i \vec{\mathbf{r}}_i}{\sum m_i}$, $\quad \vec{\mathbf{r}}_{CG}\left(\sum m_i\right) = \sum m_i \vec{\mathbf{r}}_i$

We require the center of mass to be at the origin; this simplifies the equation, leaving

$$\sum m_i x_i = 0 \quad \text{and} \quad \sum m_i y_i = 0$$

To find the x coordinate, we substitute the known values:

$$(5.00\ \text{kg})(0\ \text{m}) + (3.00\ \text{kg})(0\ \text{m}) + (4.00\ \text{kg})(3.00\ \text{m}) + (8.00\ \text{kg})x = 0$$

Solving for x gives $\quad x = -1.50\ \text{m}$.

Likewise, to find the y coordinate, we solve:

$$(5.00\ \text{kg})(0\ \text{m}) + (3.00\ \text{kg})(4.00\ \text{m}) + (4.00\ \text{kg})(0\ \text{m}) + (8.00\ \text{kg})y = 0$$

to find $\quad y = -1.50\ \text{m}$

Therefore, a fourth mass of 8.00 kg should be located at

$$\vec{\mathbf{r}}_4 = \boxed{(-1.50\hat{\mathbf{i}} - 1.50\hat{\mathbf{j}})\ \text{m}}$$

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P12.5 Let σ represent the mass-per-face area. (It would be equal to the material's density multiplied by the constant thickness of the wood.) A vertical strip at position x, with width dx and height $\dfrac{(x-3.00)^2}{9}$, has mass

$$dm = \frac{\sigma (x-3.00)^2\, dx}{9}$$

The total mass is

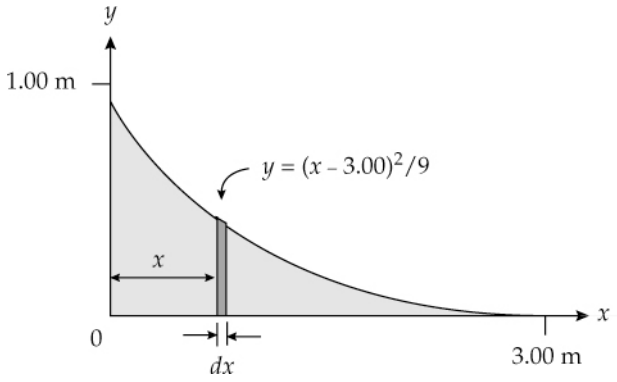
$$M = \int dm = \int_{x=0}^{3.00} \frac{\sigma (x-3)^2\, dx}{9} = \left(\frac{\sigma}{9}\right) \int_0^{3.00} \left(x^2 - 6x + 9\right) dx$$

$$= \left(\frac{\sigma}{9}\right) \left[\frac{x^3}{3} - \frac{6x^2}{2} + 9x\right]_0^{3.00} = \sigma$$

The x coordinate of the center of gravity is

$$x_{\text{CG}} = \frac{\int x\, dm}{M} = \frac{1}{9\sigma} \int_0^{3.00} \sigma x (x-3)^2\, dx = \frac{\sigma}{9\sigma} \int_0^{3.00} \left(x^3 - 6x^2 + 9x\right) dx$$

$$= \frac{1}{9} \left[\frac{x^4}{4} - \frac{6x^3}{3} + \frac{9x^2}{2}\right]_0^{3.00} = \frac{6.75 \text{ m}}{9.00} = \boxed{0.750 \text{ m}}$$



ANS. FIG. P12.5

P12.6 We can visualize this as a whole pizza with mass m_1 and center of gravity located at x_1, plus a hole that has negative mass, $-m_2$, with center of gravity at x_2:

$$x_{\text{CG}} = \frac{m_1 x_1 - m_2 x_2}{m_1 - m_2}$$

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Call σ the mass of each unit of pizza area.

$$x_{CG} = \frac{\sigma\pi R^2 0 - \sigma\pi\left(\frac{R}{2}\right)^2\left(\frac{-R}{2}\right)}{\sigma\pi R^2 - \sigma\pi\left(\frac{R}{2}\right)^2}$$

$$x_{CG} = \frac{R/8}{3/4} = \boxed{\frac{R}{6}}$$

P12.7 In a uniform gravitational field, the center of mass and center of gravity of an object coincide. Thus, the center of gravity of the triangle is located at $x = 6.67$ m, $y = 2.33$ m (see Example 9.12 on the center of mass of a triangle in Chapter 9).

The coordinates of the center of gravity of the three-object system are then:

$$x_{CG} = \frac{\sum m_i x_i}{\sum m_i}$$

$$= \frac{(6.00\text{ kg})(5.50\text{ m}) + (3.00\text{ kg})(6.67\text{ m}) + (5.00\text{ kg})(-3.50\text{ m})}{(6.00 + 3.00 + 5.00)\text{ kg}}$$

$$= \frac{35.5\text{ kg}\cdot\text{m}}{14.0\text{ kg}} = \boxed{2.54\text{ m}} \text{ and}$$

$$y_{CG} = \frac{\sum m_i y_i}{\sum m_i}$$

$$= \frac{(6.00\text{ kg})(7.00\text{ m}) + (3.00\text{ kg})(2.33\text{ m}) + (5.00\text{ kg})(+3.50\text{ m})}{14.0\text{ kg}}$$

$$= \frac{66.5\text{ kg}\cdot\text{m}}{14.0\text{ kg}} = \boxed{4.75\text{ m}}$$

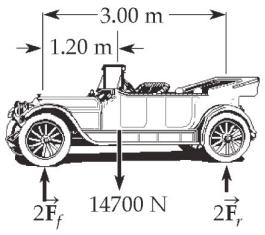
Section 12.3 Examples of Rigid Objects in Static Equilibrium

P12.8 The car's weight is

$$F_g = mg = (1\,500\text{ kg})(9.80\text{ m/s}^2)$$

$$= 1\,4700\text{ N}$$

Call $\vec{\mathbf{F}}$ the force of the ground on each of the front wheels and $\vec{\mathbf{R}}$ the normal force on each of the rear wheels. If we take torques around the front axle, with counterclockwise in the picture



ANS. FIG. P12.8

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chosen as positive, the equations are as follows:

$$\sum F_x = 0: \quad 0 = 0$$
$$\sum F_y = 0: \quad 2R - 14\ 700\text{N} + 2F = 0$$
$$\sum \tau = 0: \quad +2R(3.00\text{m}) - (14\ 700\ \text{N})(1.20\text{m}) + 2F(0) = 0$$

The torque equation gives:

$$R = \frac{17\ 640\ \text{N} \cdot \text{m}}{6.00\ \text{m}} = 2\ 940\ \text{N} = \boxed{2.94\ \text{kN}}$$

Then, from the second force equation,

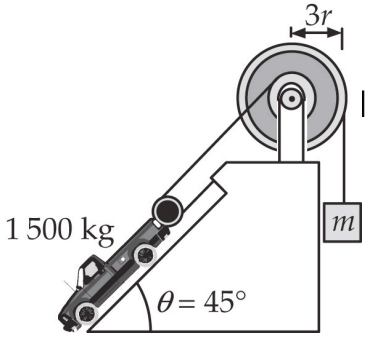
$$2(2.94\ \text{kN}) - 14.7\ \text{kN} + 2F = 0 \qquad \text{and} \qquad F = \boxed{4.41\ \text{kN}}$$

P12.9 The second condition for equilibrium at the pulley is

$$\sum \tau = 0 = mg(3r) - Tr$$

and from equilibrium at the truck, we obtain

$$2T - Mg\sin 45.0° = 0$$
$$T = \frac{Mg\sin 45.0°}{2}$$
$$= \frac{(1\ 500\ \text{kg})g\sin 45.0°}{2}$$
$$= 530g\ \text{N}$$



ANS. FIG. P12.9

solving for the mass of the counterweight from [1] and substituting gives

$$m = \frac{T}{3g} = \frac{530g}{3g} = \boxed{177\ \text{kg}}$$

P12.10 (a) For rotational equilibrium of the lowest rod about its point of support, $\sum \tau = 0$.

$$+(12.0\ \text{g})g(3.00\ \text{cm}) - m_1 g(4.00\ \text{cm}) = 0$$

which gives

$$\boxed{m_1 = 9.00\ \text{g}}$$

(b) For the middle rod,

$$+m_2 g(2.00\ \text{cm}) - (12.0\ \text{g} + 9.0\ \text{g})g(5.00\ \text{cm}) = 0$$

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which gives

$$\boxed{m_2 = 52.5 \text{ g}}$$

(c) For the top rod,

$$(52.5 \text{ g} + 12.0 \text{ g} + 9.0 \text{ g})g(4.00 \text{ cm}) - m_3 g(6.00 \text{ cm}) = 0$$

which gives

$$\boxed{m_3 = 49.0 \text{ g}}$$

P12.11 Since the beam is in equilibrium, we choose the center as our pivot point and require that

$$\sum \tau_{\text{center}} = -F_{\text{Sam}}(2.80 \text{ m}) + F_{\text{Joe}}(1.80 \text{ m}) = 0$$

or

$$F_{\text{Joe}} = 1.56 F_{\text{Sam}} \qquad \textbf{[1]}$$

Also,

$$\sum F_y = 0 \quad \Rightarrow \quad F_{\text{Sam}} + F_{\text{Joe}} = 450 \text{ N} \qquad \textbf{[2]}$$

Substitute equation [1] into [2] to get the following:

$$F_{\text{Sam}} + 1.56 F_{\text{Sam}} = 450 \text{ N} \quad \text{or} \quad F_{\text{Sam}} = \frac{450 \text{ N}}{2.56} = 176 \text{ N}$$

Then, equation [1] yields $F_{\text{Joe}} = 1.56(176 \text{ N}) = 274 \text{ N}$

$$\boxed{\begin{array}{l}\text{Sam exerts an upward force of 176 N.} \\ \text{Joe exerts an upward force of 274 N.}\end{array}}$$

P12.12 (a) To find U, measure distances and forces from point A. Then, balancing torques,

$$(0.750 \text{ m})U = (29.4 \text{ N})(2.25) \quad \boxed{U = 88.2 \text{ N}}$$

(b) To find D, measure distances and forces from point B. Then, balancing torques,

$$(0.750 \text{ m})D = (1.50 \text{ m})(29.4 \text{ N}) \quad \boxed{D = 58.8 \text{ N}}$$

Also, notice that $U = D + F_g$, so $\sum F_y = 0$.

P12.13 (a) The wall is frictionless, but it does exert a horizontal normal force, n_w.

$$\sum F_x = f - n_w = 0$$

$$\sum F_y = n_g - 800 \text{ N} - 500 \text{ N} = 0$$

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Taking torques about an axis at the foot of the ladder,

$$(800\text{ N})(4.00\text{ m})\sin 30.0^\circ + (500\text{ N})(7.50\text{ m})\sin 30.0^\circ - n_w(15.0\text{ cm})\cos 30.0^\circ = 0$$

Solving the torque equation,

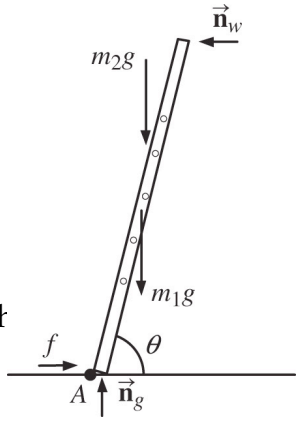
$$n_w = \frac{\left[(4.00\text{ m})(800\text{ N}) + (7.50\text{ m})(500\text{ N})\right]\tan 30.0^\circ}{15.0\text{ m}} = 268\text{ N}$$

Next substitute this value into the F_x equation to find

$f = n_w = \boxed{268\text{ N}}$ in the positive x direction.

Solving the equation $\sum F_y = 0$,

$n_g = \boxed{1\,300\text{ N}}$ in the positive y direction



ANS. FIG. P12.13(b)

(b) Refer to ANS. FIG. P12.13(b) on the right. In th equation $\sum \tau_A = 0$ gives:

$$(9.00\text{ m})(800\text{ N})\sin 30.0^\circ + (7.50\text{ m})(500\text{ N})\sin 30.0^\circ - (15.0\text{ m})(n_w)\sin 60.0^\circ = 0$$

or $n_w = 421\text{ N}$

Since $f = n_w = 421\text{ N}$ and $f = f_{\max} = \mu n_g$, we find

$$\mu = \frac{f_{\max}}{n_g} = \frac{421\text{ N}}{1\,300\text{ N}} = \boxed{0.324}$$

P12.14 (a) The wall is frictionless, but it does exert a horizontal normal force, n_w.

$$\sum F_x = f - n_w = 0 \qquad \textbf{[1]}$$

$$\sum F_y = n_g - m_1g - m_2g = 0 \qquad \textbf{[2]}$$

$$\sum \tau_A = -m_1g\left(\frac{L}{2}\right)\cos\theta - m_2gx\cos\theta + n_wL\sin\theta = 0$$

From the torque equation,

$$n_w = \left[\frac{1}{2}m_1g + \left(\frac{x}{L}\right)m_2g\right]\cot\theta$$

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Then, from equation [1]: $f = n_w = \boxed{\left[\frac{1}{2}m_1 g + \left(\frac{x}{L}\right)m_2 g\right]\cot\theta}$

and from equation [2]: $n_g = \boxed{(m_1 + m_2)g}$

(b) Refer to ANS. FIG. P12.13(b) above. If the ladder is on the verge of slipping when $x = d$, then

$$\mu = \frac{f|_{x=d}}{n_g} = \boxed{\frac{(m_1/2 + m_2 d/L)\cot\theta}{m_1 + m_2}}$$

P12.15 (a) Vertical forces on one-half of the chain are

$$T_e \sin 42.0° = 20.0 \text{ N}$$

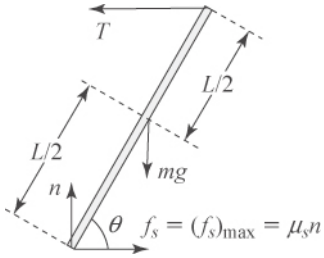
$$\boxed{T_e = 29.9 \text{ N}}$$

(b) Horizontal forces on one-half of the chain are

$$T_e \cos 42.0° = T_m$$

$$\boxed{T_m = 22.2 \text{ N}}$$

P12.16 (a) See the force diagram shown in ANS. FIG. P12.16.



ANS. FIG. P12.16

(b) Select a pivot point where an unknown force acts so that the force has no torque about that point. Picking the lower end of the beam eliminates torque from the normal force, n, and the friction force, f.

$\sum \tau_{\text{lower end}} = 0$:

$$0 + 0 - mg\left(\frac{L}{2}\cos\theta\right) + T(L\sin\theta) = 0$$

or

$$T = \frac{mg}{2}\left(\frac{\cos\theta}{\sin\theta}\right) = \boxed{\frac{mg}{2}\cot\theta}$$

(c) From the first condition for equilibrium,

$$\sum F_x = 0 \quad \Rightarrow \quad -T + \mu_s n = 0 \text{ or } T = \mu_s n \qquad \textbf{[1]}$$

$$\sum F_y = 0 \quad \Rightarrow \quad n - mg = 0 \text{ or } n = mg \qquad \textbf{[2]}$$

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Substitute equation [2] into [1] to obtain $\boxed{T = \mu_s mg}$.

(d) Equate the results of parts (b) and (c) to obtain $\boxed{\mu_s = \frac{1}{2}\cot\theta}$.

This result is valid only at the critical angle θ where the beam is on the verge of slipping (i.e., where $f_s = (f_s)_{\max}$ is valid).

(e) $\boxed{\text{The ladder slips.}}$ When the base of the ladder is moved to the left, the angle θ decreases. According to the result in part (b), the tension T increases. This requires a larger friction force to balance T, but the static friction force is already at its maximum value in ANS. FIG. P12.16.

P12.17 (a) In Figure P12.17, let the "Single point of contact" be point P, the force the nail exerts on the hammer claws be R, the mass of the hammer (1.00 kg) be M, and the normal force exerted on the hammer at point P be n, while the horizontal static friction exerted by the surface on the hammer at P be f.

Taking moments about P,

$$(R\sin 30.0^\circ)0 + (R\cos 30.0^\circ)(5.00\text{ cm}) + Mg(0) - (150\text{ N})(30.0\text{ cm}) = 0$$

$$R = 1\,039.2\text{ N} = 1.04\text{ kN}$$

The force exerted by the hammer on the nail is equal in magnitude and opposite in direction:

$\boxed{\text{1.04 kN at 60° upward and to the right}}$

(b) From the first condition for equilibrium,

$$\sum F_x = f - R\sin 30.0^\circ + 150\text{ N} = 0 \rightarrow f = 370\text{ N}$$

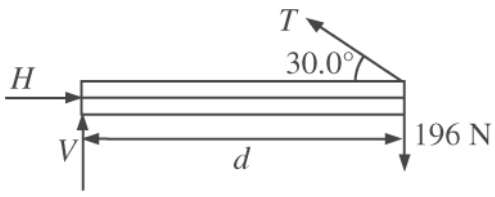
$$\sum F_y = n - Mg - R\cos 30.0^\circ = 0$$

$$\rightarrow n = (1.00\text{ kg})(9.80\text{ m/s}^2) + (1\,040\text{ N})\cos 30.0^\circ = 910\text{ N}$$

$$\boxed{\vec{\mathbf{F}}_{\text{surface}} = \left(370\hat{\mathbf{i}} + 910\hat{\mathbf{j}}\right)\text{ N}}$$

P12.18 (a) See the force diagram in ANS. FIG. P12.18.

(b) The mass M of the beam is 20.0 kg. We consider the torques acting on the beam, about an axis perpendicular



ANS. FIG. P12.18

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to the page and through the left end of the horizontal beam.

$$\sum \tau = +(T \sin 30.0^\circ)d - Mgd = 0$$

$$T = \frac{Mg}{\sin 30.0^\circ} = \frac{196 \text{ N}}{\sin 30.0^\circ} = \boxed{392 \text{ N}}$$

(c) From $\sum F_x = 0,\ H - T\cos 30.0^\circ = 0,$

or $H = (392 \text{ N})\cos 30.0^\circ = \boxed{339 \text{ N to the right}}$

(d) From $\sum F_y = 0,\ V + T\sin 30.0^\circ - 196 \text{ N} = 0,$

or $V = 196 \text{ N} - (392 \text{ N})\sin 30.0^\circ = \boxed{0}$

(e) From the same free-body diagram with the axis chosen at the right-hand end, we write

$$\sum \tau = H(0) - Vd + T(0) + 196\text{N}(0) = 0, \quad \text{so} \quad \boxed{V = 0}$$

(f) From $\sum F_y = 0,\ V + T\sin 30.0^\circ - 196 \text{ N} = 0,$

or $T = 0 + 196 \text{ N}/\sin 30.0^\circ = \boxed{392 \text{ N}}$

(g) From $\sum F_x = 0,\ H - T\cos 30.0^\circ = 0,$

or $H = (392 \text{ N})\cos 30.0^\circ = \boxed{339 \text{ N to the right}}$

(h) The two solutions agree precisely. They are equally accurate.

P12.19 The bridge has mass $M = 2\,000$ kg and the knight and horse have mass $m = 1\,000$ kg. Relative to the hinge end of the bridge, the cable is attached horizontally out a distance $x = (5.00 \text{ m})\cos 20.0^\circ = 4.70 \text{ m}$ and vertically up a distance $y = (5.00 \text{ m})\sin 20.0^\circ = 1.71 \text{ m}$. The cable then makes the following angle with the vertical wall:

$$\theta = \tan^{-1}\left[\frac{(4.70) \text{ m}}{12.0 - 1.71 \text{ m}}\right] = 24.5^\circ$$

Call the force components at the hinge H_x (to the right) and H_y (upward).

(a) Take torques about the hinge end of the bridge:

$$H_x(0) + H_y(0) - Mg(4.00 \text{ m})\cos 20.0^\circ - (T\sin 24.5^\circ)(1.71 \text{ m}) + (T\cos 24.5^\circ)(4.70 \text{ m}) - mg(7.00 \text{ m})\cos 20.0^\circ = 0$$

which yields $T = \boxed{27.7 \text{ kN}}$

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(b) $\sum F_x = 0 \Rightarrow H_x - T\sin 24.5° = 0,$

or $H_x = (27.7\text{ kN})\sin 24.5° = \boxed{11.5\text{ kN (right)}}$

(c) $\sum F_y = 0 \Rightarrow H_y - Mg + T\cos 24.5° - mg = 0$

Thus,

$$H_y = (M+m)g - (27.7\text{ kN})\cos 24.5° = -4.19\text{ kN}$$

$$= \boxed{4.19\text{ kN down}}$$

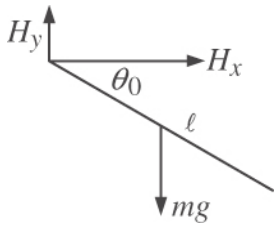
P12.20 (a) No time interval. The horse's feet lose contact with the drawbridge as soon as it begins to move.

From the result of (b) below, the tangential acceleration of the point where the horse stands is

$$a_t = \alpha r = (1.73\text{ rad/s}^2)(7.00\text{ m}) = 12.1\text{ m/s}^2$$

which has a vertical component $a_t \cos 20.0° = 11.4\text{ m/s}^2$, greater than the acceleration of gravity.

(b) Assuming that the bridge does fall from under the horse, its angular acceleration will be caused by torque from the weight of the bridge—if the bridge does not fall out from under the horse, there will be additional torque from the weight of the knight and horse, and the acceleration will be greater.



ANS. FIG. P12.20(b)

$$\sum \tau = I\alpha$$

$$Mg\left(\frac{\ell}{2}\right)\cos\theta_0 = \frac{1}{3}M\ell^2\alpha \rightarrow \alpha = \frac{3g\cos 20.0^\circ}{2(8.00\text{ m})} = \boxed{1.73\text{ rad/s}}$$

As cited in part (a), this results in the bridge falling out from under the horse, so our assumption was justified.

(c) Because there is no friction at the hinge, the bridge-Earth system is isolated, so mechanical energy is conserved. When the bridge strikes the wall:

$$K_i + U_i = K_f + U_f$$

$$Mgh = \frac{1}{2}I\omega^2 \rightarrow Mg\left(\frac{\ell}{2}\right)(1+\sin 20.0^\circ) = \frac{1}{2}\left(\frac{1}{3}M\ell^2\right)\omega^2$$

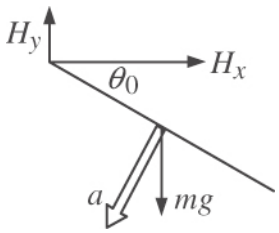
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which gives

$$\omega = \sqrt{\frac{3g\left(1+\sin 20.0^\circ\right)}{8.00 \text{ m}}} = \boxed{2.22 \text{ rad/s}}$$

(d) The tangential acceleration of the center of mass of the bridge is

$$a_t = \frac{\ell}{2}\alpha = \frac{1}{2}(8.0 \text{ m})\left(1.73 \text{ rad/s}^2\right)$$
$$= 6.92 \text{ m/s}^2$$



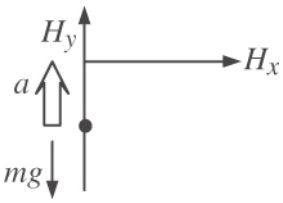
ANS. FIG. P12.20(d)

which is directed 20.0° below the horizontal. By Newton's second law:

$$\sum F_x = Ma_x$$
$$H_x = \left(2\ 000 \text{ kg}\right)\left(6.92 \text{ m/s}^2\right)\sin 20.0^\circ$$
$$= 4.72 \text{ kN}$$
$$\sum F_y = Ma_y$$
$$H_y - Mg = Ma_y$$
$$H_y = \left(2\ 000 \text{ kg}\right)\left(9.80 \text{ m/s}^2\right)$$
$$+\left(2\ 000 \text{ kg}\right)\left(-6.92 \text{ m/s}^2\right)\cos 20.0^\circ$$
$$= \boxed{6.62 \text{ kN}}$$

$\boxed{\text{The force at the hinge is } \left(4.72\hat{\mathbf{i}} + 6.62\hat{\mathbf{j}}\right) \text{ kN.}}$

(e) When the bridge strikes the wall, $H_x = 0$ and the hinge supplies a vertical centripetal force:

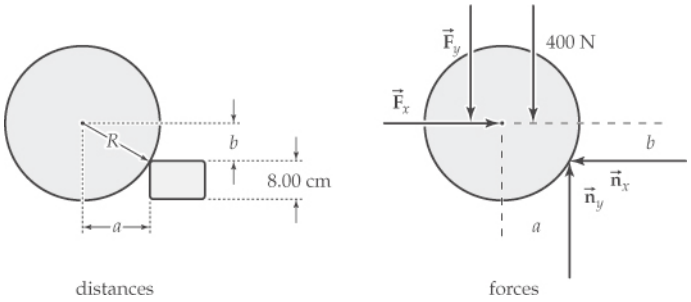


ANS. FIG. P12.20(e)

$$\sum F_y = Ma_y$$
$$H_y - Mg = Ma_y = M\omega^2\frac{\ell}{2}$$
$$H_y = Mg + M\omega^2\frac{\ell}{2} = M\left(g + \omega^2\frac{\ell}{2}\right)$$
$$H_y = \left(2\ 000 \text{ kg}\right)\left(9.80 \text{ m/s}^2 + (2.22 \text{ rad/s})^2\frac{8.00 \text{ m}}{2}\right)$$
$$H_y = \boxed{59.1 \text{ kJ}}$$

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P12.21 Call the required force F, with components $F_x = F\cos 15.0°$ and $F_y = -F\sin 15.0°$, transmitted to the center of the wheel by the handles.



ANS. FIG. P12.21

Just as the wheel leaves the ground, the ground exerts no force on it.

$$\sum F_x = 0: \quad F\cos 15.0° - n_x = 0 \qquad \textbf{[1]}$$

$$\sum F_y = 0: \quad -F\sin 15.0° - 400\text{ N} + n_y = 0 \qquad \textbf{[2]}$$

Take torques about its contact point with the brick. The needed distances are seen to be:

$$b = R - 8.00\text{ cm} = (20.0 - 8.00)\text{ cm} = 12.0\text{ cm}$$

$$a = \sqrt{R^2 - b^2} = \sqrt{(20.0\text{ cm})^2 - (8.00\text{ cm})^2} = 16.0\text{ cm}$$

(a) $\sum \tau = 0: \quad -F_x b + F_y a + (400\text{ N})a = 0,$ or

$$F\left[-(12.0\text{ cm})\cos 15.0° + (16.0\text{ cm})\sin 15.0°\right] + (400\text{ N})(16.0\text{ cm}) = 0$$

so $$F = \frac{6\,400\text{ N}\cdot\text{cm}}{7.45\text{ cm}} = \boxed{859\text{ N}}$$

(b) Then, using equations [1] and [2],

$$n_x = (859\text{ N})\cos 15.0° = 830\text{ N} \text{ and}$$

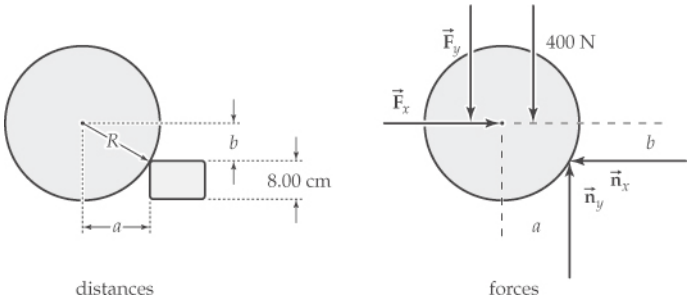
$$n_y = 400\text{ N} + (859\text{ N})\sin 15.0° = 622\text{ N}$$

$$n = \sqrt{n_x^2 + n_y^2} = \boxed{1.04\text{ kN}}$$

$$\theta = \tan^{-1}\left(\frac{n_y}{n_x}\right) = \tan^{-1}(0.749) = \boxed{36.9°\text{ to the left and upward}}$$

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P12.22 Call the required force F, with components $F_x = F\cos\theta$ and $F_y = -F\sin\theta$, transmitted to the center of the wheel by the handles.



ANS. FIG. P12.22

Just as the wheel leaves the ground, the ground exerts no force on it.

$$\sum F_x = 0: \quad F\cos\theta - n_x = 0 \qquad \textbf{[1]}$$

$$\sum F_y = 0: \quad -F\sin\theta - mg + n_y = 0 \qquad \textbf{[2]}$$

Take torques about its contact point with the brick. The needed distances are seen to be:

$$b = R - h$$

$$a = \sqrt{R^2 - (R-h)^2} = \sqrt{2Rh - h^2}$$

(a) $\sum\tau = 0: \quad -F_x b + F_y a + mga = 0,$ or

$$F\left[-b\cos\theta + a\sin\theta\right] + mga = 0$$

$$\rightarrow F = \frac{mga}{b\cos\theta - a\sin\theta} = \boxed{\frac{mg\sqrt{2Rh-h^2}}{(R-h)\cos\theta - \sqrt{2Rh-h^2}\,\sin\theta}}$$

(b) Then, using equations [1] and [2],

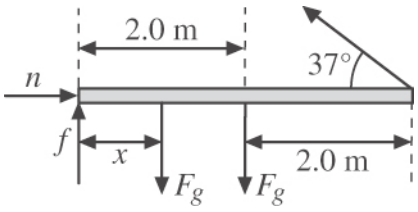
$$n_x = F\cos\theta = \boxed{\frac{mg\sqrt{2Rh-h^2}\,\cos\theta}{(R-h)\cos\theta - \sqrt{2Rh-h^2}\,\sin\theta}}$$

and $n_y = F\sin\theta + mg = \boxed{mg\left[1 + \frac{\sqrt{2Rh-h^2}\,\cos\theta}{(R-h)\cos\theta - \sqrt{2Rh-h^2}\,\sin\theta}\right]}$

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P12.23 When $x = x_{\min}$, the rod is on the verge of slipping, so

$$f = (f_s)_{\max} = \mu_s n = 0.50n$$



ANS. FIG. P12.23

From $\sum F_x = 0,\ n - T\cos 37° = 0$

or $n = 0.799T$

Thus, $f = 0.50(0.799T) = 0.399T$

From $\sum F_y = 0,\ f + T\sin 37° - 2F_g = 0,$

or $0.399T + 0.602T - 2F_g = 0,$ giving $T = 2.00F_g$

Using $\sum \tau = 0$ for an axis perpendicular to the page and through the left end of the beam gives

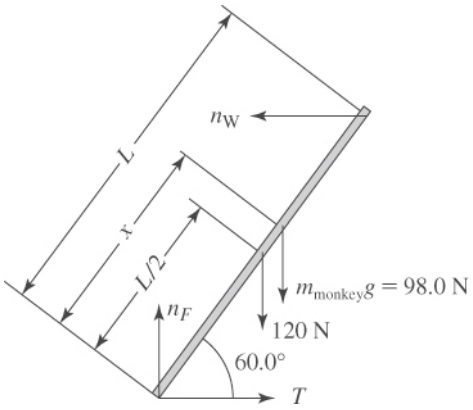
$$-F_g \cdot x_{\min} - F_g(2.0\text{ m}) + \left[(2F_g)\sin 37°\right](4.0\text{ m}) = 0$$

which reduces to $x_{\min} =$ $\boxed{2.81\text{ m}}$

P12.24 (a) The force diagram is shown in ANS. FIG. P12.24.

(b) From $\sum F_y = 0 \Rightarrow n_F - 120\text{ N} - m_{\text{monkey}}g = 0$

$$n_F = 120\text{ N} + (10.0\text{ kg})(9.80\text{ m/s}^2) = \boxed{218\text{ N}}$$



ANS. FIG. P12.24

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(c) When $x = 2L/3$, we consider the bottom end of the ladder as our pivot and obtain

$$\sum \tau\big|_{\substack{\text{bottom}\\ \text{end}}} = 0:$$

$$-(120\text{ N})\left(\frac{\not{L}}{2}\cos 60.0^\circ\right) - (98.0\text{ N})\left(\frac{2\not{L}}{3}\cos 60.0^\circ\right) + n_W\left(\not{L}\sin 60.0^\circ\right) = 0$$

$$\text{or}\quad n_W = \frac{\left[60.0\text{ N} + (196/3)\text{ N}\right]\cos 60.0^\circ}{\sin 60.0^\circ} = 72.4\text{ N}$$

Then, $\sum F_x = 0 \Rightarrow T - n_W = 0$ or $T = n_W = \boxed{72.4\text{ N}}$

(d) When the rope is ready to break, $T = n_W = 80.0$ N. Then $\sum \tau\big|_{\substack{\text{bottom}\\ \text{end}}} = 0$ yields

$$-(120\text{ N})\left(\frac{L}{2}\cos 60.0^\circ\right) - (98.0\text{ N})x\cos 60.0^\circ + (80.0\text{ N})(L\sin 60.0^\circ) = 0$$

$$\text{or}\quad x = \frac{\left[(80.0\text{ N})\sin 60.0^\circ - (60.0\text{ N})\cos 60.0^\circ\right]L}{(98.0\text{ N})\cos 60.0^\circ}$$

$$= 0.802L = 0.802(3.00\text{ m}) = \boxed{2.41\text{ m}}$$

(e) If the horizontal surface were rough and the rope removed, a horizontal static friction force directed toward the wall would act on the bottom end of the ladder. Otherwise, the analysis would be much as what is done above. The maximum distance the monkey could climb would correspond to the condition that the friction force have its maximum value, $\mu_s n_F$, so you would need to know the coefficient of static friction between the ladder and the floor to solve part (d).

P12.25 Consider the torques about an axis perpendicular to the page and through the left end of the plank. $\sum \tau = 0$ gives

$$-(700\text{ N})(0.500\text{ m}) - (294\text{ N})(1.00\text{ m}) + \left(T_1\sin 40.0^\circ\right)(2.00\text{ m}) = 0$$

$$\text{or}\quad T_1 = \boxed{501\text{ N}}$$

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Then, $\sum F_x = 0$ gives

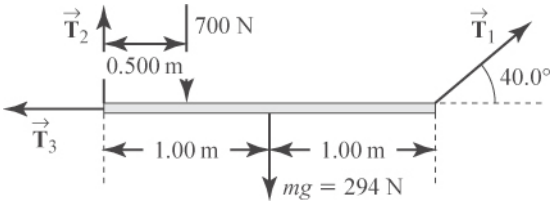
$$-T_3 + T_1 \cos 40.0° = 0$$

or $$T_3 = (501\text{ N})\cos 40.0° = \boxed{384\text{ N}}$$

From $\sum F_y = 0$,

$$T_2 - 994\text{ N} + T_1 \sin 40.0° = 0,$$

or $$T_2 = 994\text{ N} - (501\text{ N})\sin 40.0° = \boxed{672\text{ N}}$$



ANS. FIG. P12.25

Section 12.4 Elastic Properties of Solids

P12.26 Count the wires. If they are wrapped together so that all support nearly equal stress, the number should be

$$\frac{20.0\text{ kN}}{0.200\text{ kN}} = 100$$

Since cross-sectional area is proportional to diameter squared, the diameter of the cable will be

$$(1\text{ mm})\sqrt{100}\ \boxed{\sim 1\text{ cm}}$$

P12.27 We use $B = -\dfrac{\Delta P}{\Delta V / V_i} = -\dfrac{\Delta P V_i}{\Delta V}$.

(a) $$\Delta V = -\frac{\Delta P V_i}{B} = -\frac{(1.13\times 10^8\ \text{N/m}^2)(1\text{ m}^3)}{0.21\times 10^{10}\ \text{N/m}^2} = \boxed{-0.053\,8\text{ m}^3}$$

(b) The quantity of water with mass 1.03×10^3 kg occupies volume at the bottom: $1\text{ m}^3 - 0.053\,8\text{ m}^3 = 0.946\text{ m}^3$.

So its density is $\dfrac{1.03\times 10^3\text{ kg}}{0.946\text{ m}^3} = \boxed{1.09\times 10^3\ \text{kg/m}^3}$

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(c) With only a 5% volume change in this extreme case, liquid water is indeed nearly incompressible.

P12.28 (a) We find the maximum force from the equation for stress:

$$\text{stress} = \frac{F}{A} = \frac{F}{\pi r^2}$$

$$F = (\text{stress})\pi\left(\frac{d}{2}\right)^2$$

$$F = \left(1.50\times10^{8}\ \text{N/m}^2\right)\pi\left(\frac{2.50\times10^{-2}\ \text{m}}{2}\right)^2$$

$$F = \boxed{73.6\ \text{kN}}$$

(b) From the definition of Young's modulus,

$$\text{stress} = Y(\text{strain}) = \frac{Y\Delta L}{L_i}$$

$$\Delta L = \frac{(\text{stress})L_i}{Y} = \frac{\left(1.50\times10^{8}\ \text{N/m}^2\right)(0.250\ \text{m})}{1.50\times10^{10}\ \text{N/m}^2} = \boxed{2.50\ \text{mm}}$$

P12.29 From the defining equation for the shear modulus, we find Δx as

$$\Delta x = \frac{hf}{SA} = \frac{\left(5.00\times10^{-3}\ \text{m}\right)\left(20.0\ \text{N}\right)}{\left(3.0\times10^{6}\ \text{N/m}^2\right)\left(14.0\times10^{-4}\ \text{m}^2\right)} = 2.38\times10^{-5}\,\text{m}$$

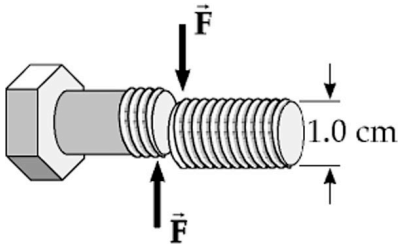
or $$\Delta x = \boxed{2.38\times10^{-2}\ \text{mm}}$$

P12.30 The definition of Young's modulus, $Y = \dfrac{\text{stress}}{\text{strain}}$, means that Y is the slope of the graph:

$$Y = \frac{300\times10^{6}\ \text{N/m}^2}{0.003} = \boxed{1.0\times10^{11}\ \text{N/m}^2}$$

P12.31 (a) From ANS. FIG. P12.31(a),

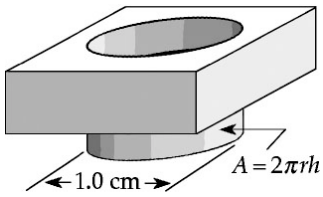
$$F = \sigma A$$
$$= \left(4.00\times10^{8}\ \text{N/m}\right)$$
$$\times\left[\pi\left(0.500\times10^{-2}\ \text{m}\right)^2\right]$$
$$= \boxed{3.14\times10^{4}\ \text{N}}$$



ANS. FIG. P12.31(a)

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(b) Now the area of the molecular layers sliding over each other is the curved lateral surface area of the cylinder being punched out, a cylinder of radius 0.500 cm and height 0.500 cm. So,



ANS. FIG. P12.31(b)

$$F = \sigma A$$
$$= \sigma(h)(2\pi r)$$
$$= \left(4.00 \times 10^{8}\ \text{N/m}\right)(2\pi)\left(0.500 \times 10^{-2}\ \text{m}\right) \times \left(0.500 \times 10^{-2}\ \text{m}\right)$$
$$= \boxed{6.28 \times 10^{4}\ \text{N}}$$

P12.32 Let V represent the original volume. Then, $0.090\ 0V$ is the change in volume that would occur if the block cracked open. Imagine squeezing the ice, with unstressed volume $1.09V$, back down to its previous volume, so $\Delta V = -0.090\ 0V$. According to the definition of the bulk modulus as given in the chapter text, we have

$$\Delta P = -\frac{B(\Delta V)}{V_i}$$
$$= -\frac{\left(2.00 \times 10^{9}\ \text{N/m}^2\right)(-0.090\ 0V)}{1.09V}$$
$$= \boxed{1.65 \times 10^{8}\ \text{N/m}^2}$$

P12.33 Young's Modulus is given by $Y = \dfrac{F/A}{\Delta L/L_i}$.

The load force is

$$F = (200\ \text{kg})(9.80\ \text{m/s}^2) = 1\ 960\ \text{N}.$$

so
$$\Delta L = \frac{FL_i}{AY} = \frac{(1\ 960\ \text{N})(4.00\ \text{m})(1\ 000\ \text{mm/m})}{\left(0.200 \times 10^{-4}\ \text{m}^2\right)\left(8.00 \times 10^{10}\ \text{N/m}^2\right)} = \boxed{4.90\ \text{mm}}$$

P12.34 Part of the load force extends the cable and part compresses the column by the same distance $\Delta\ell$:

$$F = \frac{Y_A A_A \Delta\ell}{\ell_A} + \frac{Y_s A_s \Delta\ell}{\ell_s}$$

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from which we obtain

$$\Delta \ell = \frac{F}{Y_A A_A / \ell_A + Y_s A_s / \ell_s}$$

$$= \frac{8\,500 \text{ N}}{\left(7 \times 10^{10}\right)\pi\left(0.162\,4^2 - 0.161\,4^2\right)/4(3.25) + 20 \times 10^{10}\pi\left(0.012\,7\right)^2/4(5.75)}$$

$$= \boxed{8.60 \times 10^{-4} \text{ m}}$$

P12.35 Let the 3.00-kg mass be mass #1, with the 5.00-kg mass, mass # 2. Applying Newton's second law to each mass gives

$$m_1 a = T - m_1 g \qquad \textbf{[1]}$$

and $$m_2 a = m_2 g - T \qquad \textbf{[2]}$$

where T is the tension in the wire.

Solving equation [1] for the acceleration gives

$$a = \frac{T}{m_1} - g$$

and substituting this into equation [2] yields

$$\frac{m_2}{m_1} T - m_2 g = m_2 g - T$$

Solving for the tension T gives

$$T = \frac{2 m_1 m_2 g}{m_2 + m_1} = \frac{2\left(3.00 \text{ kg}\right)\left(5.00 \text{ kg}\right)\left(9.80 \text{ m/s}^2\right)}{8.00 \text{ kg}} = 36.8 \text{ N}$$

From the definition of Young's modulus, $Y = \dfrac{F L_i}{A\left(\Delta L\right)}$, the elongation of the wire is:

$$\Delta L = \frac{T L_i}{Y A} = \frac{(36.8 \text{ N})(2.00 \text{ m})}{\left(2.00 \times 10^{11} \text{ N/m}^2\right)\pi\left(2.00 \times 10^{-3} \text{ m}\right)^2}$$

$$= \boxed{0.029\,2 \text{ mm}}$$

P12.36 A particle under a net force model:

$$\left|\overline{F}\right| = \frac{m\left|v_f - v_i\right|}{\Delta t}$$

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Hence,

$$\left|\bar{F}\right| = \frac{30.0\ \text{kg}\left|-10.0\ \text{m/s} - 20.0\ \text{m/s}\right|}{0.110\ \text{s}} = 8.18 \times 10^3\ \text{N}$$

By Newton's third law, this is also the magnitude of the average force exerted on the spike by the hammer during the blow. Thus, the stress in the spike is

$$\text{Stress} = \frac{F}{A} = \frac{8.18 \times 10^3\ \text{N}}{\pi\left(0.023\,0\ \text{m}\right)^2 / 4} = 1.97 \times 10^7\ \text{N/m}^2$$

and the strain is

$$\text{strain} = \frac{\text{stress}}{Y} = \frac{1.97 \times 10^7\ \text{N/m}^2}{20.0 \times 10^{10}\ \text{N/m}^2} = \boxed{9.85 \times 10^{-5}}$$

Additional Problems

P12.37 Let n_A and n_B be the normal forces at the points of support. Then, from the translational equilibrium equation in the y direction, we have

$$\sum F_y = 0: \quad n_A + n_B - \left(8.00 \times 10^4\ \text{kg}\right)g - \left(3.00 \times 10^4\ \text{kg}\right)g = 0$$

Choosing the axis at point A, we find, from the condition for rotational equilibrium:

$$\sum \tau = 0:$$
$$-\left(3.00 \times 10^4\ \text{kg}\right)(15.0\ \text{m})g - \left(8.00 \times 10^4\ \text{kg}\right)(25.0\ \text{m})g + n_B(50.0\ \text{m}) = 0$$

We can solve the torque equation directly to find

$$n_B = \frac{\left[\left(3.00 \times 10^4\ \text{kg}\right)(15.0\ \text{m}) + \left(8.00 \times 10^4\ \text{kg}\right)(25.0\ \text{m})\right](9.80\ \text{m/s}^2)}{50.0\ \text{m}}$$
$$= 4.80 \times 10^5\ \text{N}$$

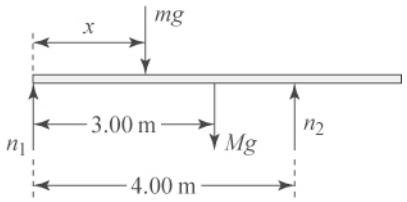
Then the force equation gives

$$n_A = (8.00 \times 10^4\ \text{kg} + 3.00 \times 10^4\ \text{kg})(9.80\ \text{m/s}^2) - 4.80 \times 10^5\ \text{N}$$
$$= 5.98 \times 10^5\ \text{N}$$

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P12.38 (a) Rigid object in static equilibrium.

(b) ANS. FIG. P12.38 shows the free-body diagram.



ANS. FIG. P12.38

$Mg = (90.0\text{ kg})g = 882\text{ N}$, and $mg = (55.0\text{ kg})g = 539\text{ N}$.

(c) Note that about the right pivot, only n_1 exerts a clockwise torque, all other forces exert counterclockwise torques except for n_2 which exerts zero torque. The woman is at $x = 0$ when n_1 is greatest. With this location of the woman, the counterclockwise torque about the center of the beam is a maximum. Thus, n_1 must be exerting its maximum clockwise torque about the center to hold the beam in rotational equilibrium.

(d) $n_1 = 0$ As the woman walks to the right along the beam, she will eventually reach a point where the beam will start to rotate clockwise about the rightmost pivot. At this point, the beam is starting to lift up off of the leftmost pivot and the normal force exerted by that pivot will have diminished to zero.

(e) When the beam is about to tip, $n_1 = 0$, and

$$\sum F_y = 0 \text{ gives } 0 + n_2 - Mg - mg = 0, \text{ or}$$

$$n_2 = Mg + mg = 882\text{ N} + 539\text{ N} = \boxed{1.42 \times 10^3\text{ N}}$$

(f) Requiring that the net torque be zero about the right pivot when the beam is about to tip ($n_1 = 0$) gives

$$\sum \tau = n_2(0) + (4.00\text{ m} - x)mg + (4.00\text{ m} - 3.00\text{ m})Mg = 0$$

or $$(mg)x = (1.00\text{ m})Mg + (4.00\text{ m})mg, \text{ and}$$

$$x = (1.00\text{ m})\frac{M}{m} + 4.00\text{ m}$$

Thus, $$x = (1.00\text{ m})\frac{(90.0\text{ kg})}{(55.0\text{ kg})} + 4.00\text{ m} = \boxed{5.64\text{ m}}$$

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(g) When $n_1 = 0$ and $n_2 = 1.42 \times 10^3$ N, for torque about the left pivot:

$$\sum \tau = 0 - (539\text{ N})x - (882\text{ N})(3.00\text{ m}) + \left(1.42 \times 10^3\text{ N}\right)(4.00\text{ m}) = 0$$

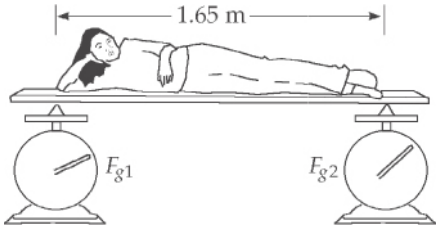
or $$x = \frac{-3.03 \times 10^3\text{ N} \cdot \text{m}}{-539\text{ N}} = \boxed{5.62\text{ m}}$$

which, within limits of rounding errors, is

the same as the answer to part (f).

P12.39 $\sum F_y = 0$: $+380\text{ N} - F_g + 320\text{ N} = 0$

$F_g = 700$ N



ANS. FIG. P12.39

Take torques about her feet:

$$\sum \tau = 0: \quad -380\text{ N}(1.65\text{ m}) + (700\text{ N})x + (320\text{ N})0 = 0$$

$x =$ 0.896 m

P12.40 When the concrete has cured and the pre-stressing tension has been released, the rod presses in on the concrete and with equal force, T_2, the concrete produces tension in the rod.

(a) In the concrete:

$$\text{stress} = 8.00 \times 10^6\text{ N/m}^2 = Y \cdot (\text{strain}) = Y\left(\frac{\Delta L}{L_i}\right)$$

Thus,

$$\Delta L = \frac{(\text{stress})L_i}{Y} = \frac{\left(8.00 \times 10^6\text{ N/m}^2\right)(1.50\text{ m})}{30.0 \times 10^9\text{ N/m}^2}$$

or $\Delta L = 4.00 \times 10^{-4}$ m $=$ 0.400 mm

(b) In the concrete:

$$\text{stress} = \frac{T_2}{A_c} = 8.00 \times 10^6\text{ N/m}^2$$

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so

$$T_2 = \left(8.00\times10^{6}\ \text{N/m}^2\right)\left(50.0\times10^{-4}\ \text{m}^2\right) = \boxed{40.0\ \text{kN}}$$

(c) For the rod:

$$\frac{T_2}{A_R} = \left(\frac{\Delta L}{L_i}\right)Y_{\text{steel}} \quad \text{so} \quad \Delta L = \frac{T_2 L_i}{A_R Y_{\text{steel}}}$$

$$\Delta L = \frac{\left(4.00\times10^{4}\ \text{N}\right)(1.50\ \text{m})}{\left(1.50\times10^{-4}\ \text{m}^2\right)\left(20.0\times10^{10}\ \text{N/m}^2\right)}$$

$$= 2.00\times10^{-3}\ \text{m} = \boxed{2.00\ \text{mm}}$$

(d) The rod in the finished concrete is 2.00 mm longer than its unstretched length. To remove stress from the concrete, one must stretch the rod 0.400 mm farther, by a total of $\boxed{2.40\ \text{mm}}$.

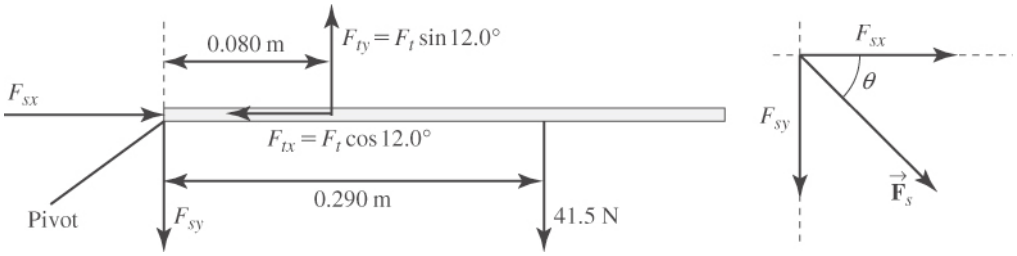
(e) For the stretched rod around which the concrete is poured:

$$\frac{T_1}{A_R} = \left(\frac{\Delta L_{\text{total}}}{L_i}\right)Y_{\text{steel}} \quad \text{or} \quad T_1 = \left(\frac{\Delta L_{\text{total}}}{L_i}\right)A_R Y_{\text{steel}}$$

$$T_1 = \left(\frac{2.40\times10^{-3}\ \text{m}}{1.50\ \text{m}}\right)\left(1.50\times10^{-4}\ \text{m}^2\right)\left(20.0\times10^{10}\ \text{N/m}^2\right)$$

$$= \boxed{48.0\ \text{kN}}$$

P12.41 We reproduce the forces in ANS. FIG. P12.41.



ANS. FIG. P12.41

Requiring that $\sum\tau = 0$, using the shoulder joint at point O as a pivot, gives

$$\sum\tau = \left(F_t \sin 12.0°\right)(0.080\ 0\ \text{m}) - (41.5\ \text{N})(0.290\ \text{m}) = 0$$

or $F_t = \boxed{724\ \text{N}}$

Then

$$\sum F_y = 0 \quad \Rightarrow \quad -F_{sy} + (724\ \text{N})\sin 12.0° - 41.5\ \text{N} = 0$$

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yielding $F_{sy} = 109\ \text{N}$

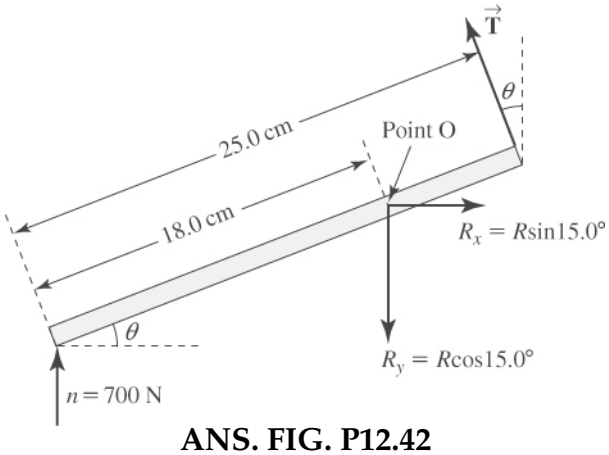
$\sum F_x = 0$ then gives

$$F_{sx} - (724\ \text{N})\cos 12.0° = 0, \text{ or } F_{sx} = 708\ \text{N}$$

Therefore,

$$F_s = \sqrt{F_{sx}^2 + F_{sy}^2} = \sqrt{(708\ \text{N})^2 + (109\ \text{N})^2} = \boxed{716\ \text{N}}$$

P12.42 In the free-body diagram of the foot given at the right, note that the force $\vec{\mathbf{R}}$ (exerted on the foot by the tibia) has been replaced by its horizontal and vertical components. Employing both conditions of equilibrium (using point O as the pivot point) gives the following three equations:



ANS. FIG. P12.42

$$\sum F_x = 0 \Rightarrow R\sin 15.0° - T\sin\theta = 0$$

or $$R = \frac{T\sin\theta}{\sin 15.0°} \quad \textbf{[1]}$$

$$\sum F_y = 0 \Rightarrow 700\ \text{N} - R\cos 15.0° + T\cos\theta = 0 \quad \textbf{[2]}$$

$$\sum \tau_O = 0 \Rightarrow -(700\ \text{N})[(18.0\ \text{cm})\cos\theta] + T(25.0\ \text{cm} - 18.0\ \text{cm}) = 0$$

or $$T = (1\ 800\ \text{N})\cos\theta \quad \textbf{[3]}$$

Substituting equation [3] into equation [1] gives

$$R = \left(\frac{1\ 800\ \text{N}}{\sin 15.0°}\right)\sin\theta\cos\theta \quad \textbf{[4]}$$

Substituting equations [3] and [4] into equation [2] yields

$$\left(\frac{1\ 800\ \text{N}}{\tan 15.0°}\right)\sin\theta\cos\theta - (1\ 800\ \text{N})\cos^2\theta = 700\ \text{N}$$

which reduces to: $\sin\theta\cos\theta = (\tan 15.0°)\cos^2\theta + 0.104\ 2$

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Squaring this result and using the identity $\sin^2\theta = 1 - \cos^2\theta$ gives

$$\left[\tan^2(15.0°) + 1\right]\cos^4\theta + \left[(2\tan 15.0°)(0.104\,2) - 1\right]\cos^2\theta + (0.104\,2)^2 = 0$$

In this last result, let $u = \cos^2\theta$ and evaluate the constants to obtain the quadratic equation:

$$(1.071\,8)u^2 - (0.944\,2)u + 0.010\,9 = 0$$

The quadratic formula yields the solutions u = 0.869 3 and u = 0.0117. Thus,

$$\theta = \cos^{-1}\left(\sqrt{0.869\,3}\right) = 21.2° \quad \text{or} \quad \theta = \cos^{-1}\left(\sqrt{0.011\,7}\right) = 83.8°$$

We ignore the second solution since it is physically impossible for the human foot to stand with the sole inclined at 83.8° to the floor. We are the left with $\theta = \boxed{21.2°}$.

Equation [3] then yields

$$T = (1\,800\text{ N})\cos 21.2° = \boxed{1.68\text{ kN}}$$

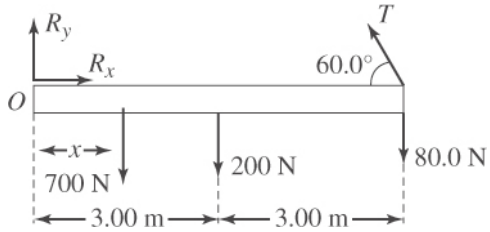
and equation [1] gives

$$R = \frac{(1.68 \times 10^3\text{ N})\sin 21.2°}{\sin 15.0°} = \boxed{2.34\text{ kN}}$$

P12.43 (a) ANS. FIG. P12.43 shows the force diagram.

(b) If x = 1.00 m, then

$$\sum \tau_O = (-700\text{ N})(1.00\text{ m}) - (200\text{ N})(3.00\text{ m}) - (80.0\text{ N})(6.00\text{ m}) + (T\sin 60.0°)(6.00\text{ m}) = 0$$



ANS. FIG. P12.43

Solving for the tension gives: $T = \boxed{343\text{ N}}$.

From $\sum F_x = 0$, $R_x = T\cos 60.0° = \boxed{171\text{ N}}$.

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From $\sum F_y = 0$, $R_y = 980\text{ N} - T\sin 60.0° =$ $\boxed{683\text{ N}}$.

(c) If $T = 900$ N:

$$\sum \tau_O = (-700\text{ N})x - (200\text{ N})(3.00\text{ m}) - (80.0\text{ N})(6.00\text{ m}) + [(900\text{ N})\sin 60.0°](6.00\text{ m}) = 0$$

Solving for x gives $x =$ $\boxed{5.14\text{ m}}$.

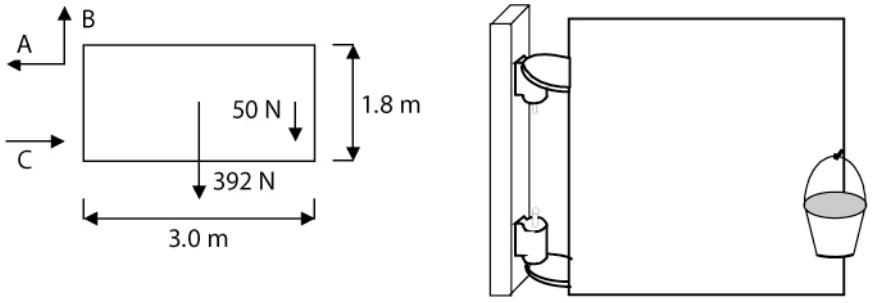
P12.44 (a) See ANS. FIG. P12.44 for the force diagram. See the solution in the textbook. The weight of the uniform gate is 392 N. It is 3.00 m wide. The hinges are separated vertically by 1.80 m. The bucket of grain weighs 50.0 N. One of the hinges, which we suppose is the upper one, supports the whole weight of the gate. Find the components of the forces that both hinges exert on the gate.

(b) From the torque equation,

$$C = \frac{738\text{ N}\cdot\text{m}}{1.8\text{ m}} = 410\text{ N}$$

Then $A = 410$ N. Also $B = 442$ N.

The upper hinge exerts 410 N to the left and 442 N up.
The lower hinge exerts 410 N to the right.



ANS. FIG. P12.44

P12.45 We know that the direction of the force from the cable at the right end is along the cable, at an angle of θ above the horizontal. On the other end, we do not know magnitude or direction for the hinge force $\vec{\mathbf{R}}$ so we show it as two unknown components.

The first condition for equilibrium gives two equations:

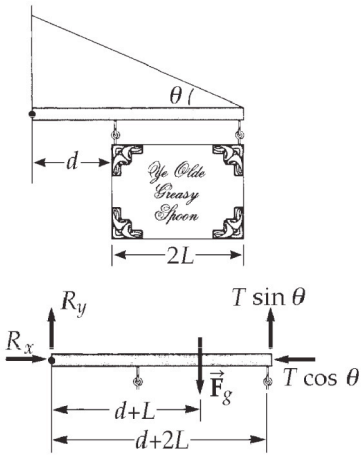
$$\sum F_x = 0: \quad +R_s - T\cos\theta = 0$$
$$\sum F_y = 0: \quad +R_y - F_g + T\sin\theta = 0$$

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Taking torques about the left end, we find the second condition is

$$\sum \tau = 0:$$

$$R_y(0) + R_s(0) - F_g(d+L) + (0)(T\cos\theta) + (d+2L)(T\sin\theta) = 0$$



ANS. FIG. P12.45

(a) The torque equation gives $T = \boxed{\dfrac{F_g(L+d)}{\sin\theta(2L+d)}}$

(b) Now from the force equations,

$$R_x = \boxed{\frac{F_g(L+d)\cot\theta}{2L+d}} \quad \text{and} \quad R_y = \boxed{\frac{F_g L}{2L+d}}$$

P12.46 ANS. FIG. P12.46 shows the force diagram.

$\sum \tau_{\text{point O}} = 0$ gives

$$(T\cos 25.0^\circ)\left(\frac{3\ell}{4}\sin 65.0^\circ\right) + (T\sin 25.0^\circ)\left(\frac{3\ell}{4}\cos 65.0^\circ\right)$$

$$= (2\,000\text{ N})(\ell\cos 65.0^\circ) + (1\,200\text{ N})\left(\frac{\ell}{2}\cos 65.0^\circ\right)$$

From which, $T = 1\,465\text{ N} = \boxed{1.46\text{ kN}}$

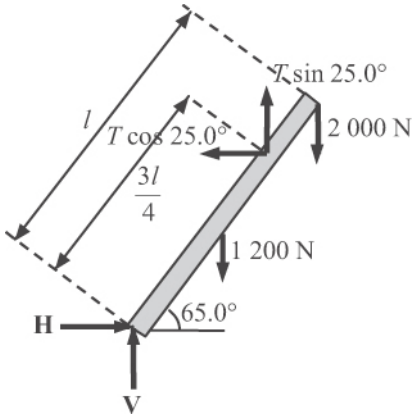
From $\sum F_x = 0$,

$$H = T\cos 25.0^\circ = 1\,328\text{ N}(\text{toward right}) = \boxed{1.33\text{ kN}}$$

From $\sum F_y = 0$,

$$V = 3\,200\text{ N} - T\sin 25.0^\circ = 2\,581\text{ N}(\text{upward}) = \boxed{2.58\text{ kN}}$$

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ANS. FIG. P12.46

P12.47 We interpret the problem to mean that the support at point B is frictionless. Then the support exerts a force in the x direction and

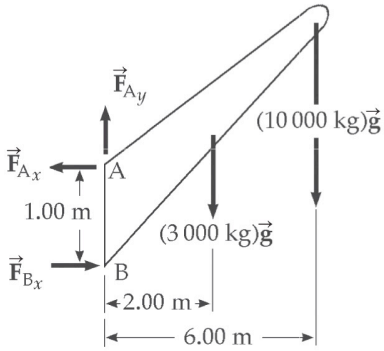
$$F_{By} = 0$$

$$\sum F_x = F_{Bx} - F_{Ax} = 0$$

$$F_{Ay} - (3\,000 + 10\,000)g = 0$$

and

$$\sum \tau = -(3\,000g)(2.00) - (10\,000g)(6.00) + F_{Bx}(1.00) = 0$$



ANS. FIG. P12.47

These equations combine to give the magnitudes of the components:

$$F_{Ax} = F_{Bx} = 6.47 \times 10^5 \text{ N}$$

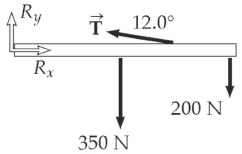
$$F_{Ay} = 1.27 \times 10^5 \text{ N}$$

The forces are:	$\vec{\mathbf{F}}_A = (-6.47 \times 10^5 \hat{\mathbf{i}} + 1.27 \times 10^5 \hat{\mathbf{j}})$ N
and	$\vec{\mathbf{F}}_B = 6.47 \times 10^5 \hat{\mathbf{i}}$ N

P12.48 (a) Choosing torques about the hip joint, $\sum \tau = 0$ gives

$$-\frac{L}{2}(350 \text{ N}) + (T \sin 12.0°)\left(\frac{2L}{3}\right) - (200 \text{ N})L = 0$$

From which, $T =$ 2.71 kN .



ANS. FIG. P12.48

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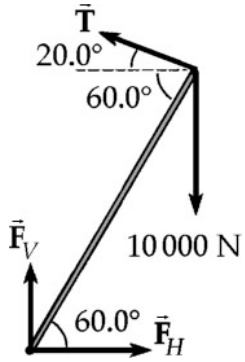
(b) Let R_x = compression force along spine, and from $\sum F_x = 0$:

$$R_x = T_x = T\cos 12.0° = \boxed{2.65 \text{ kN}}$$

(c) You should lift "with your knees" rather than "with your back." In this situation, with a load weighing only 200 N, you can make the compressional force in your spine about ten times smaller by bending your knees and lifting with your back as straight as possible.

(d) In this situation, you can make the compressional force in your spine about ten times smaller by bending your knees and lifting with your back as straight as possible

P12.49 From ANS. FIG. P12.49, the angle $\vec{\mathbf{T}}$ makes with the rod is θ = 60.0°+ 20.0° = 80.0° and the perpendicular component of $\vec{\mathbf{T}}$ is T sin 80.0°.



ANS. FIG. P12.49

Summing torques around the base of the rod, and applying Newton's second law in the horizontal and vertical directions, we have

$$\sum \tau = 0: \quad -(4.00 \text{ m})(10\,000 \text{ N})\cos 60° + T(4.00 \text{ m})\sin 80.0° = 0$$

(a) Solving the above equation for T gives

$$T = \frac{(10\,000 \text{ N})\cos(60.0°)}{\sin(80.0°)} = \boxed{5.08 \text{ kN}}$$

(b) In the horizontal direction,

$$\sum F_x = 0: \quad F_H - T\cos(20.0°) = 0$$

so $F_H = T\cos(20.0°) = 4.77$ kN

(c) From $\sum F_y = 0: \quad F_V + T\sin(20.0°) - 10\,000 \text{ N} = 0$,

we find

$$F_V = (10\,000 \text{ N}) - T\sin(20.0°) = \boxed{8.26 \text{ kN}}$$

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P12.50 The cabinet has height $\ell = 1.00$ m, width $w = 0.600$ m, and weight $Mg = 400$ N. The force $F = 300$ N is applied by the worker in the first case at height $h_1 = 0.100$ m and in the second at height $h_2 = 0.650$ m.

Consider the magnitudes of the torques about the lower right front edge of the cabinet from the weight Mg and from the applied force F for the two values of h.

The torque from the weight is the same in each case:

Cases 1 and 2

$$\tau_G = Mg\frac{w}{2}$$
$$= (400\text{ N})(0.300\text{ m}) = 120\text{ N}\cdot\text{m}$$

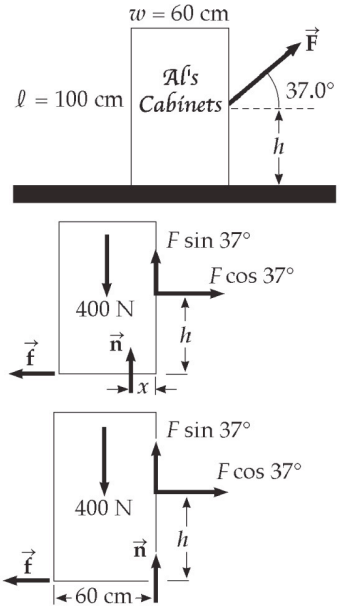
The torque from force F is different in each case:

Case 1

$$\tau_F = (F\cos 37.0°)h_1$$
$$= (300\text{ N}\cos 37.0°)(0.100\text{ m})$$
$$\tau_F = 24.0\text{ N}\cdot\text{m}$$

Case 2

$$\tau_F = (F\cos 37.0°)h_2$$
$$= (300\text{ N}\cos 37.0°)(0.650\text{ m})$$
$$\tau_F = 156\text{ N}\cdot\text{m}$$



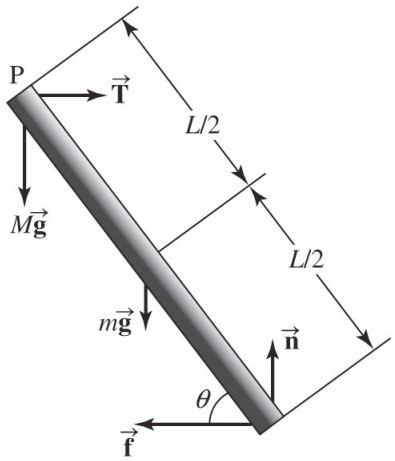
ANS. FIG. P12.50

We see in Case 1 that the counterclockwise torque from the weight is greater than the clockwise torque from the applied force. If the cabinet is to slide without acceleration, the net torque must be zero; this is possible because the normal force from the floor can provide additional clockwise torque. We see in Case 2, however, that the counterclockwise torque from the weight is smaller than the clockwise torque from the applied force, but no other force is available to provide addition counterclockwise torque, so the net torque cannot be zero.

The situation is impossible because the new technique would tip the cabinet over.

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P12.51 (a) We use $\sum F_x = \sum F_y = \sum \tau = 0$ and choose the axis at the point of contact with the floor to simplify the torque analysis. Since the rope is described as very rough, we will assume that it will never slip on the end of the beam. First, let us determine what friction force at the floor is necessary to put the system in equilibrium; then we can check whether that friction force can be obtained.



ANS. FIG. P12.51

$$\sum F_x = 0: \quad T - f = 0$$
$$\sum F_y = 0: \quad n - Mg - mg = 0$$
$$\sum \tau = 0: \quad Mg(\cos\theta)L + mg(\cos\theta)\frac{L}{2} - T(\sin\theta)L = 0$$

Solving the torque equation, we find $T = \left(M + \frac{1}{2}m\right) g \cot\theta.$

Then the horizontal-force equation implies by substitution that this same expression is equal to f. In order for the beam not to slip, we need $f \le \mu_s n.$ Substituting for n and f from the above equations, we obtain the requirement

$$\mu_s \ge \left[\frac{M + m/2}{M + m}\right]\cot\theta$$

The factor in brackets is always < 1, so if $\mu \ge \cot\theta$ then M can be increased without limit. In this case, there is no maximum mass! Otherwise, if $\mu_s < \cot\theta,$ the equality will apply on the verge of slipping, and solving for M yields

$$M = \boxed{\frac{m}{2}\left[\frac{2\mu_s \sin\theta - \cos\theta}{\cos\theta - \mu_s \sin\theta}\right]}$$

(b) At the floor, we see that the normal force is in the y direction and the friction force is in the $-x$ direction. The reaction force exerted by the floor then has magnitude

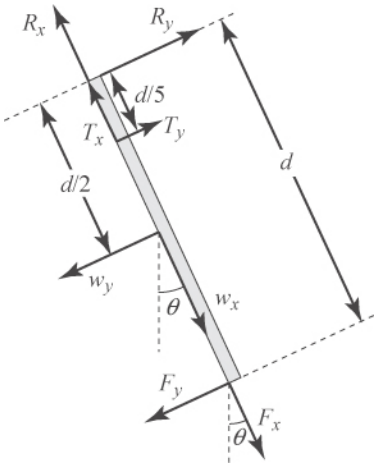
$$R = \sqrt{n^2 + (\mu_s n)^2} = \boxed{g(M + m)\sqrt{1 + \mu_s^2}}$$

(c) At point P, the force of the beam on the rope is in magnitude

$$F = \sqrt{T^2 + (Mg)^2} = \boxed{g\sqrt{M^2 + \mu_s^2 (M + m)^2}}$$

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P12.52 First, we resolve all forces into components parallel to and perpendicular to the tibia, as shown. Note that $\theta = 40.0°$ and



ANS. FIG. P12.52

$$w_y = (30.0\text{ N})\sin 40.0° = 19.3\text{ N}$$
$$F_y = (12.5\text{ N})\sin 40.0° = 8.03\text{ N}$$

and

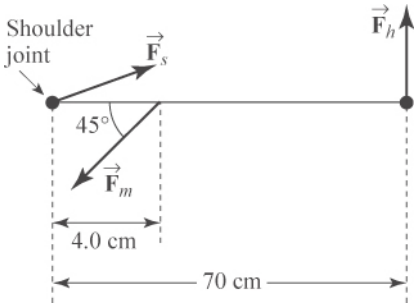
$$T_y = T\sin 25.0°$$

Using $\sum\tau = 0$ for an axis perpendicular to the page and through the upper end of the tibia gives

$$(T\sin 25.0°)\frac{d}{5} - (19.3\text{ N})\frac{d}{2} - (8.03\text{ N})d = 0$$

or $T = \boxed{209\text{ N}}$

P12.53 (a) From the symmetry of the situation, we may conclude that the magnitude of the upward force on each hand is half the weight of the athlete: $F_h = 750\text{ N}/2 = 375\text{ N}$. Considering the shoulder joint as the pivot, the second condition of equilibrium gives



ANS. FIG. P12.53

$$\sum\tau = 0 \;\Rightarrow\; F_h(70.0\text{ cm}) - (F_m\sin 45°)(4.00\text{ cm}) = 0$$

or $$F_m = \frac{(375\text{ N})(70.0\text{ cm})}{(4.00\text{ cm})\sin 45°} = \boxed{9.28\text{ kN}}$$

(b) The moment arm of the force is no longer 70.0 cm from the shoulder joint but only 49.5 cm:

$$\sum\tau = 0 \;\Rightarrow\; F_h(70.0\text{ cm})\sin 45° - (F_m\sin 45°)(4.00\text{ cm}) = 0$$
$$F_m = \frac{(375\text{ N})(70.0\text{ cm})}{(4.00\text{ cm})} = \boxed{6.56\text{ kN}}$$

therefore reducing F_m to 6.56 kN.

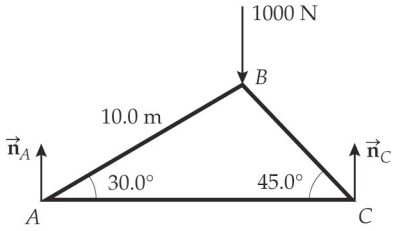
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P12.54 (a) The height of pin B is

$$(10.0\text{ m})\sin 30.0^\circ = 5.00\text{ m}$$

The length of bar BC is then

$$\overline{BC} = \frac{5.00\text{ m}}{\sin 45.0^\circ} = 7.07\text{ m}$$



ANS. FIG. P12.54(a)

Consider the entire truss:

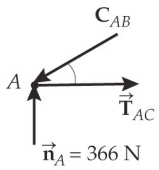
$$\sum F_y = n_A - 1\,000\text{ N} + n_C = 0$$

$$\sum \tau_A = -(1\,000\text{ N})10.0\cos 30.0^\circ + n_C[10.0\cos 30.0^\circ + 7.07\cos 45.0^\circ] = 0$$

Which gives $\boxed{n_C = 634\text{ N}}$.

Then, $\boxed{n_A = 1\,000\text{ N} - n_C = 366\text{ N}}$

(b) Joint A: $\sum F_y = 0$: $-C_{AB}\sin 30.0^\circ + 366\text{ N} = 0$

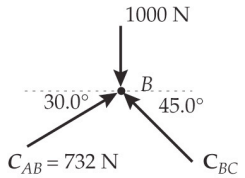


so $C_{AB} = \boxed{732\text{ N}}$

$\sum F_x = 0$:

$$-C_{AB}\cos 30.0^\circ + T_{AC} = 0$$

$$T_{AC} = (732\text{ N})\cos 30.0^\circ = \boxed{634\text{ N}}$$



ANS. FIG. P12.54(b)

Joint B:

$$\sum F_x = 0: \quad (732\text{ N})\cos 30.0^\circ - C_{BC}\cos 45.0^\circ = 0$$

$$C_{BC} = \frac{(732\text{ N})\cos 30.0^\circ}{\cos 45.0^\circ} = \boxed{897\text{ N}}$$

P12.55 Considering the torques about the point at the bottom of the bracket yields:

$$W(0.0500\text{ m}) - F_{\text{hor}}(0.0600\text{ m}) = 0 \text{ so } F_{\text{hor}} = 0.833W$$

(a) With $W = 80.0$ N, $F_{\text{hor}} = 0.833(80\text{ N}) = \boxed{66.7\text{ N}}$.

(b) Differentiate with respect to time: $dF_{\text{hor}}/dt = 0.833\,dW/dt$.

Given that $dW/dt = 0.150$ N/s:

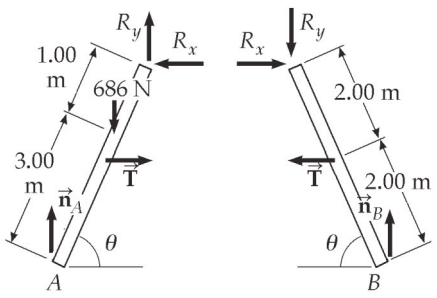
The force exerted by the screw is increasing at the rate $dF_{\text{hor}}/dt = 0.833(0.150\text{ N/s}) = 0.125\text{ N/s}$.

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P12.56 Refer to the solution to P12.57 for a general discussion of the solution.

From the geometry of the ladder, observe that

$$\cos\theta = \frac{1}{4} \rightarrow \theta = 75.5°$$



ANS. FIG. P12.56

In the following, we use the variables $m = 70.0$ kg, length $AC = BC = \ell = 4.00$ m, and $d = 3.00$ m.

Consider the net torque about point A (on the bottom left side of the ladder) from external forces on the whole ladder. The torques about A come from the weight of the painter and the normal force n_B.

$$\sum \tau_A = -mgd\cos 75.5° + n_B\frac{\ell}{2} = 0$$

$$\rightarrow n_B = \frac{2}{\ell}mgd\cos 75.5° = \frac{2}{\ell}mgd\left(\frac{1}{4}\right) \rightarrow n_B = \frac{mgd}{2\ell}$$

Consider the net torque about point B (on the bottom right side of the ladder) from external forces on the whole ladder. The torques about B come from the weight of the painter and the normal force n_A.

$$\sum \tau_B = -n_A\frac{\ell}{2} + mg\left(\frac{\ell}{2} - d\cos 75.5°\right) = 0$$

$$\rightarrow n_A\frac{\ell}{2} = mg\left(\frac{\ell}{2} - d\cos 75.5°\right)$$

$$n_A = \frac{2}{\ell}mg\left(\frac{\ell}{2} - d\cos 75.5°\right) = mg\left(1 - \frac{d}{2\ell}\right)$$

Consider the torque from external forces about point C at the top of the right half of the ladder:

$$\sum \tau_C = -T\frac{\ell}{2}\sin 75.5° + n_B\frac{\ell}{4} = 0$$

$$\rightarrow T = n_B\frac{1}{2\sin 75.5°} = \frac{mgd}{2\ell}\frac{1}{2\sin 75.5°}$$

$$\rightarrow T = \frac{mgd}{4\ell\sin 75.5°}$$

Note that the tension T on the right half of the ladder must pull to the left, otherwise it could not contribute a clockwise torque about C to balance the counterclockwise torque from n_B.

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Now we find the components of the reaction force that the left half of the ladder exerts on the right half. Consider the forces acting on the right half of the ladder:

$$\sum F_x = R_x - T = 0 \rightarrow R_x = T, \text{ to the right}$$

$$\sum F_y = R_y + n_B = 0 \rightarrow R_y = -n_B \rightarrow R_y = n_B, \text{ downward}$$

Collecting our results, we find

(a) $$T = \frac{mgd}{4\ell \sin 75.5°} = \frac{(70.0 \text{ kg})(9.80 \text{ m/s}^2)(3.00 \text{ m})}{4(4.00 \text{ m})\sin 75.5°} \rightarrow \boxed{T = 133 \text{ N}}$$

(b) $$n_A = mg\left(1 - \frac{d}{2\ell}\right)$$

$$n_A = (70.0 \text{ kg})(9.80 \text{ m/s}^2)\left(1 - \frac{2(3.00 \text{ m})(1/4)}{4.00 \text{ m}}\right) \rightarrow \boxed{n_A = 429 \text{ N}}$$

and

$$n_B = \frac{mgd}{2\ell} = \frac{(70.0 \text{ kg})(9.80 \text{ m/s}^2)(3.00 \text{ m})}{2(4.00 \text{ m})} \rightarrow \boxed{n_B = 257 \text{ N}}$$

(c) The force exerted by the left half of the ladder on the right half is to the right and downward:

$$R_x = T \rightarrow \boxed{R_x = 133 \text{ N, to the right}}$$

and $$R_y = -n_b \rightarrow R_y = -257 \text{ N} \rightarrow \boxed{R_y = 257 \text{ N, downward}}$$

P12.57 From the geometry of Figure P12.56 and ANS. FIG. P12.56, we observe that

$$\cos\theta = \frac{\ell/4}{\ell} = \frac{1}{4}$$

and

$$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \left(\frac{1}{4}\right)^2} = \sqrt{1 - \frac{1}{16}} = \sqrt{\frac{15}{16}}$$

$$\sin\theta = \frac{\sqrt{15}}{4}$$

(a) Below in part (b) we show that normal force $n_B = mgd/2\ell$. We use this result here to find the tension T in the horizontal bar.

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Consider the torque about point C at the top of the right half of the ladder:

$$\sum \tau_C = -T\frac{\ell}{2}\sin\theta + n_B\frac{\ell}{4} = 0$$

$$T = n_B\left(\frac{\ell}{4}\right)\frac{2}{\ell\sin\theta} = \frac{mgd}{2\ell}\left(\frac{\ell}{4}\right)\frac{2}{\ell\sin\theta} = \frac{mgd}{4\ell\sin\theta} = \frac{mgd}{4\ell\left(\sqrt{15}/4\right)}$$

$$\boxed{T = \frac{mgd}{\ell\sqrt{15}}}$$

Note that the tension T on the right half of the ladder must pull to the left, otherwise it could not contribute a clockwise torque about C to balance the counterclockwise torque from n_B.

(b) We now proceed to find the normal forces n_A and n_B.

First, consider the net torque from all forces acting on the ladder about point B at the bottom right side of the whole ladder. Note that tension T on the left half of the ladder and tension T on the right half of the ladder have opposite torques because they have the same moment arms about point B, so their torques cancel (they are forces internal to the system, so they cannot contribute to net torque). In like manner, torques from R_x and R_y on both halves of the ladder cancel in pairs (again, they are internal forces). The only contributing torques come from the weight of the painter and the normal force n_A (these are forces external to the ladder).

$$\sum \tau_B = -n_A\frac{\ell}{2} + mg\left(\frac{\ell}{2} - d\cos\theta\right) = 0$$

$$n_A\frac{\ell}{2} = mg\left(\frac{\ell}{2} - \frac{d}{4}\right)$$

$$n_A\frac{\ell}{2} = mg\left(\frac{2\ell - d}{4}\right)$$

$$n_A = \frac{2}{\ell}mg\left(\frac{2\ell - d}{4}\right)$$

$$\boxed{n_A = \frac{mg(2\ell - d)}{2\ell}}$$

Now, consider the net torque from all forces acting on the ladder about point A on the bottom left side of the whole ladder. Similarly to the case of the torques about point B, the only contributing torques about A come from the weight of the painter

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and the normal force n_B (again, these are external forces).

$$\sum \tau_A = -mgd\cos\theta + n_B\frac{\ell}{2} = 0$$

$$\rightarrow n_B = \frac{2}{\ell}mgd\cos\theta = \frac{2}{\ell}mgd\left(\frac{1}{4}\right) \rightarrow \boxed{n_B = \frac{mgd}{2\ell}}$$

(c) Now we find the components of the reaction force that the left half of the ladder exerts on the right half. Consider the forces acting on the right half of the ladder:

$$\sum F_x = R_x - T = 0 \rightarrow R_x = T$$

$$\boxed{R_x = \frac{mgd}{\sqrt{15}}, \text{ to the right}}$$

$$\sum F_y = R_y + n_B = 0 \rightarrow R_y = -n_B = -\frac{mgd}{2\ell}$$

$$\boxed{R_y = \frac{mgd}{2\ell}, \text{ downward}}$$

P12.58 (a) $$F = m\left(\frac{\Delta v}{\Delta t}\right) = (1.00\text{ kg})\frac{(10.0 - 1.00)\text{ m/s}}{0.002\text{ s}} = \boxed{4\,500\text{ N}}$$

(b) $$\text{stress} = \frac{F}{A} = \frac{4\,500\text{ N}}{(0.010\text{ m})(0.100\text{ m})} = \boxed{4.50\times10^6\text{ N/m}^2}$$

(c) $\boxed{\text{Yes.}}$ This is more than sufficient to break the board.

P12.59 (a) Take both balls together. Their weight is $2mg = 3.33$ N and their CG is at their contact point.

$$\sum F_x = 0: + P_3 - P_1 = 0 \rightarrow P_3 = P_1$$

$$\sum F_y = 0: + P_2 - 2mg = 0 \rightarrow P_2 = 2mg = \boxed{3.33\text{ N}}$$

For torque about the contact point (CP) between the balls:

$$\sum \tau_{CP} = 0: \quad P_1(R\cos 45.0°) - P_2(R\cos 45.0°) + P_3(R\cos 45.0°)$$
$$- mg(R\cos 45.0°) + mg(R\cos 45.0°) = 0$$
$$\rightarrow P_1 - P_2 + P_3 = 0 \rightarrow P_1 + P_3 = P_2$$

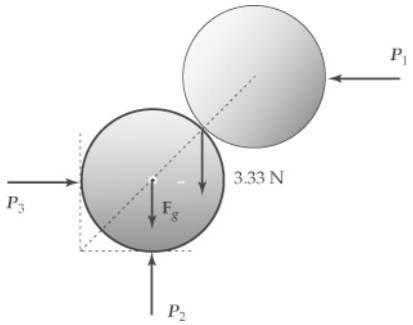
Substituting $P_3 = P_1$, we find

$$2P_1 = P_2 = 2mg \rightarrow P_1 = mg$$

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Therefore,

$$P_1 = P_3 = \boxed{1.67\ \text{N}}$$



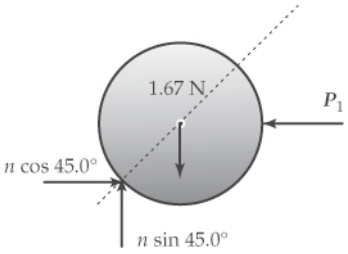
ANS. FIG. P12.59(a)

(b) Take the upper ball. The lines of action of its weight, of P_1, and of the normal force n exerted by the lower ball all go through its center, so for rotational equilibrium there can be no frictional force.

$$\sum F_x = 0: \quad n\cos 45.0° - P_1 = 0$$

$$n = \frac{1.67\ \text{N}}{\cos 45.0°} = \boxed{2.36\ \text{N}}$$

$\sum F_y = 0$: $n\sin 45.0° - 1.67\ \text{N} = 0$ gives the same result.



ANS. FIG. P12.59(b)

P12.60 We will let F represent some stretching force and use algebra to combine the Hooke's-law account of the stretching with the Young's-modulus account. Then integration will reveal the work done as the wire extends.

(a) According to Hooke's law, $|\vec{\mathbf{F}}| = k\Delta L$

Young's modulus is defined as $Y = \dfrac{F/A}{\Delta L/L}$

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By substitution,

$$Y = k\frac{L}{A} \qquad \text{or} \qquad k = \boxed{\frac{YA}{L}}$$

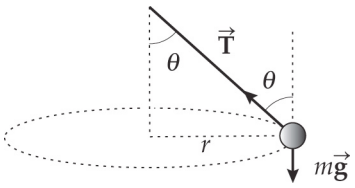
(b) The spring exerts force $-kx$. The outside agent stretching it exerts force $+kx$. We can determine the work done by integrating the force kx over the distance we stretch the wire.

$$W = -\int_0^{\Delta L} F\,dx = -\int_0^{\Delta L}(-kx)dx = \frac{YA}{L}\int_0^{\Delta L} x\,dx = \left[\frac{YA}{L}\left(\frac{1}{2}x^2\right)\right]_{x=0}^{x=\Delta L}$$

Therefore,

$$W = \boxed{\frac{1}{2}YA(\Delta L)^2 / L}$$

P12.61 Let θ represent the angle of the wire with the vertical. The radius of the circle of motion is $r = L\sin\theta$, where $L = 0.850$ m.



ANS. FIG. P12.61

For the mass:

$$\sum F_r = ma_r = m\frac{v^2}{r} = mr\omega^2$$

$$T\sin\theta = m[L\sin\theta]\omega^2$$

Further, $\frac{T}{A} = Y\cdot\frac{\Delta L}{L}$ or $T = AY\cdot\frac{\Delta L}{L}$

Thus, $AY\cdot(\Delta L/L) = mL\omega^2$, giving

$$\omega = \sqrt{\frac{AY\cdot(\Delta L/L)}{mL}} = \sqrt{\frac{\pi\left(3.90\times10^{-4}\ \text{m}\right)^2\left(7.00\times10^{10}\ \text{N/m}^2\right)\left(1.00\times10^{-3}\right)}{(1.20\ \text{kg})(0.850\ \text{m})}}$$

or $\omega = \boxed{5.73\ \text{rad/s}}$

P12.62 (a), (b) Use the first diagram and sum the torques about the lower front corner of the cabinet.

$$\sum\tau = 0 \Rightarrow -F(1.00\ \text{m}) + (400\ \text{N})(0.300\ \text{m}) = 0$$

yielding $F = \frac{(400\ \text{N})(0.300\ \text{m})}{1.00\ \text{m}} = \boxed{120\ \text{N}}$

$$\sum F_x = 0 \Rightarrow -f + 120\ \text{N} = 0, \quad \text{or} \quad f = 120\ \text{N}$$

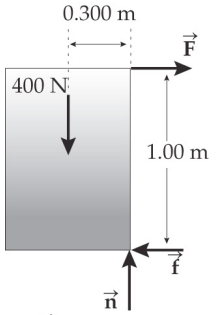
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$$\sum F_y = 0 \Rightarrow -400\text{ N} + n = 0$$

so $n = 400\text{ N}$

Thus,

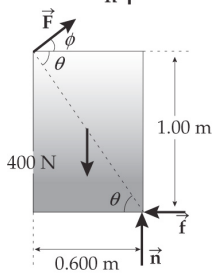
$$\mu_s = \frac{f}{n} = \frac{120\text{ N}}{400\text{ N}} = \boxed{0.300}$$



(c) Apply F' at the upper rear corner and directed so $\theta + \phi = 90.0°$ to obtain the largest possible lever arm.

$$\theta = \tan^{-1}\left(\frac{1.00\text{ m}}{0.600\text{ m}}\right) = 59.0°$$

Thus, $\phi = 90.0° - 59.0° = 31.0°$



ANS. FIG. P12.62

Sum the torques about the lower front corner of the cabinet:

$$-F'\sqrt{(1.00\text{ m})^2 + (0.600\text{ m})^2} + (400\text{ N})(0.300\text{ m}) = 0$$

so $F' = \dfrac{120\text{ N}\cdot\text{m}}{1.17\text{ m}} = 103\text{ N}$

Therefore, the minimum force required to tip the cabinet is

103 N applied at 31.0° above the horizontal at the upper left corner

***P12.63** (a) Consider the torques about an axis perpendicular to the page through the left end of the rod, as shown in ANS. FIG. P12.63.

$$\sum \tau = 0:$$

$$T(6.00\text{ m})\cos 30.0° - (100\text{ N})(3.00\text{ m}) - (500\text{ N})(4.00\text{ m}) = 0$$

then,

$$T = \frac{(100\text{ N})(3.00\text{ m}) + (500\text{ N})(4.00\text{ m})}{(6.00\text{ m})\cos 30.0°}$$

$$= \boxed{443\text{ N}}$$

(b) From the first condition for equilibrium,

$$\sum F_x = 0:$$

$$R_x = T\sin 30.0° = (443\text{ N})\sin 30.0°$$

$$= \boxed{221\text{ N toward the right}}$$

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Similarly,

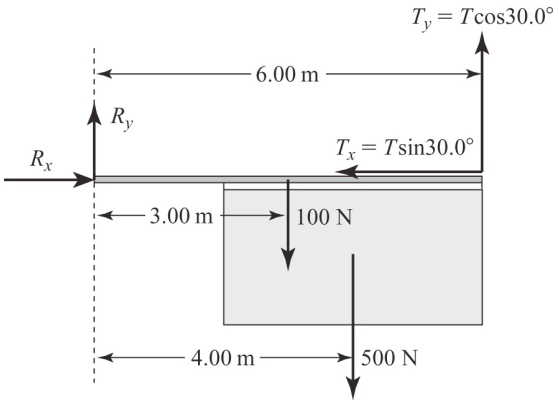
$$\sum F_y = 0:$$

$$R_y + T\cos 30.0° - 100\text{ N} - 500\text{ N} = 0$$

which gives

$$R_y = 600\text{ N} - T\cos 30.0° = 600\text{ N} - (443\text{ N})\cos 30.0°$$

$$= \boxed{217\text{ N upward}}$$

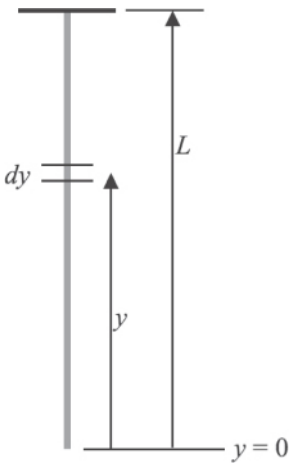


ANS. FIG. P12.63

P12.64 Let the original length (when the cable is laid horizontally on a frictionless surface) of an infinitesimal piece of the cable be dy. Let the extension of this piece be dL when the cable is hung vertically. Then, for the entire cable,

$$\Delta L = \int dL = \int \frac{F}{AY}\,dy$$

where F is the weight of the cable below a point at position y. Evaluating F, with μ as the mass per unit length,



ANS. FIG. P12.64

$$\Delta L = \int \frac{(\mu y)g}{AY}\,dy = \frac{\mu g}{AY}\int_0^L y\,dy$$

$$= \frac{\mu g}{AY}\left(\frac{L^2}{2}\right) = \frac{1}{2}\left(\frac{\mu g L^2}{AY}\right)$$

$$\Delta L = \frac{1}{2}\left[\frac{(2.40\text{ kg/m})(9.80\text{ m/s}^2)(500\text{ m})^2}{(2.00\times 10^{11}\text{ N/m}^2)(3.00\times 10^{-4}\text{ m}^2)}\right]$$

$$= 0.049\,0\text{ m} = \boxed{4.90\text{ cm}}$$

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Challenge Problems

P12.65 With ℓ as large as possible, n_1 and n_2 will both be large. The equality sign in $f_2 \le \mu_s n_2$ will be true, but the less-than sign will apply in $f_1 < \mu_s n_1$. Take torques about the lower end of the pole.

$$n_2 \ell \cos\theta + F_g\left(\frac{1}{2}\ell\right)\cos\theta - f_2 \ell \sin\theta = 0$$

Setting $f_2 = 0.576 n_2$, the torque equation becomes

$$n_2\left(1 - 0.576\tan\theta\right) + \frac{1}{2}F_g = 0$$

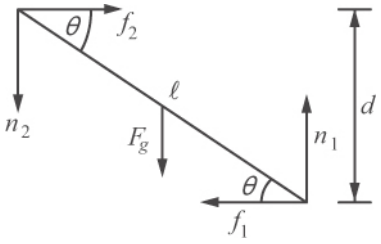
Since $n_2 > 0$, it is necessary that

$$1 - 0.576\tan\theta < 0$$

$$\therefore \tan\theta > \frac{1}{0.576} = 1.736$$

$$\therefore \theta > 60.1°$$

$$\therefore \ell = \frac{d}{\sin\theta} < \frac{7.80 \text{ ft}}{\sin 60.1°} = \boxed{9.00 \text{ ft}}$$



ANS. FIG. P12.65

P12.66 Consider forces and torques on the beam.

$$\sum F_x = 0: \quad R\cos\theta - T\cos 53° = 0$$

$$\sum F_y = 0: \quad R\sin\theta + T\sin 53° - 800 \text{ N} = 0$$

$$\sum \tau = 0: \quad (T\sin 53°)(8.00 \text{ m}) - (600 \text{ N})d - (200 \text{ N})(4.00 \text{ m}) = 0$$

(a) Suppressing units, we find

$$T = \frac{600d + 800}{8\sin 53°} = \boxed{93.9d + 125, \text{ in N}}$$

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(b) From substituting back,

$$R\cos\theta = [93.9d + 125]\cos 53.0°$$

$$R\sin\theta = 800\text{ N} - [93.9d + 125]\sin 53.0°$$

Dividing,

$$\tan\theta = \frac{R\sin\theta}{R\cos\theta} = -\tan 53.0° + \frac{800\text{ N}}{(93.9d+125)\cos 53.0°}$$

$$\boxed{\tan\theta = \left(\frac{32}{3d+4} - 1\right)\tan 53.0°}$$

(c) To find R we can work out $R^2\cos^2\theta + R^2\sin^2\theta = R^2$. From the expressions above for $R\cos\theta$ and $R\sin\theta$,

$$R^2 = T^2\cos^2 53° + T^2\sin^2 53° - 1\,600T\sin 53° + (800\text{ N})^2$$

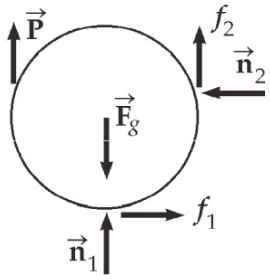
$$R^2 = T^2 - 1\,600T\sin 53° + 640\,000$$

$$R^2 = (93.9d + 125)^2 - 1\,278(93.9d + 125) + 640\,000$$

$$\boxed{R = \left(8.82\times 10^3 d^2 - 9.65\times 10^4 d + 4.96\times 10^5\right)^{1/2}}$$

(d) As d increases, T grows larger, θ decreases, and R decreases until about $d = 5.4$ m, then it increases. Notes as d increases, the d^2 term predominates.

P12.67 Imagine gradually increasing the force P. This will make the force of static friction at the bottom increase, so that the normal force at the wall increases and the friction force at the wall *can* increase. As P reaches its maximum value, the cylinder will turn clockwise microscopically to stress the welds at both contact points and make both forces of friction increase to their maximum values.



ANS. FIG. P12.67

When it is on the verge of slipping, the cylinder is in equilibrium.

$$\sum F_x = 0: \qquad \rightarrow f_1 = n_2 = \mu_s n_1 \quad \text{and} \quad f_2 = \mu_s n_2$$

$$\sum F_y = 0: \qquad \rightarrow P + n_1 + f_2 = F_g$$

$$\sum \tau = 0: \qquad \rightarrow -PR + f_1 R + f_2 R = 0 \rightarrow P = f_1 + f_2$$

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As P grows, so do f_1 and f_2. Therefore, since $\mu_s = \frac{1}{2}$,

$$f_1 = \frac{n_1}{2} \quad \text{and} \quad f_2 = \frac{n_2}{2} = \frac{n_1}{4}$$

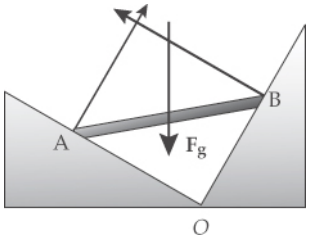
then $$P + n_1 + \frac{n_1}{4} = F_g$$

and $$P = \frac{n_1}{2} + \frac{n_1}{4} = \frac{3}{4} n_1$$

So $P + \frac{5}{4} n_1 = F_g$ becomes $P + \frac{5}{4}\left(\frac{4}{3}P\right) = F_g$ or $\frac{8}{3}P = F_g$.

Therefore, $\boxed{P = \frac{3}{8} F_g}$.

P12.68 (a) Just three forces act on the rod: forces perpendicular to the sides of the trough at A and B, and its weight. The lines of action of the normal forces at A and B will intersect at a point above the rod so that those forces will have no torque about this point. The rod's weight will cause a torque about the point of intersection as in ANS. FIG. P12.68(a), and the rod will not be in equilibrium unless the center of the rod lies vertically below the intersection point, as in ANS. FIG. P12.68(b). All three forces must be concurrent. Then the line of action of the weight is a diagonal of the rectangle formed by the two normal forces, and the rod's center of gravity is vertically above the bottom of the trough.



ANS. FIG. P12.68(a)

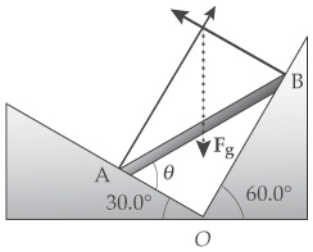
(b) In ANS. FIG. P12.68(b), $\overline{\text{AO}} \cos 30.0° = \overline{\text{BO}} \cos 60.0°$ and

$$L^2 = \overline{\text{AO}}^2 + \overline{\text{BO}}^2 = \overline{\text{AO}}^2 + \overline{\text{AO}}^2 \left(\frac{\cos^2 30.0°}{\cos^2 60.0°} \right)$$

$$\overline{\text{AO}} = \frac{L}{\sqrt{1 + \left(\frac{\cos 30°}{\cos 60°} \right)^2}} = \frac{L}{2}$$

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So $\cos\theta = \dfrac{\overline{AO}}{L} = \dfrac{1}{2}$ and $\theta = \boxed{60.0°}$



ANS. FIG. P12.68(b)

(c) Unstable. If the rod is displaced slightly, it will slip until it lies along the left edge of the trough where its center of gravity will be lower.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P12.2 The situation is impossible because x is larger than the remaining portion of the beam, which is 0.200 m long.

P12.4 $x = -1.50$ m; $y = -1.50$ m

P12.6 $\dfrac{R}{6}$

P12.10 2.94 kN; 4.41 kN

P12.8 (a) $m_1 = 9.00$ g; (b) $m_2 = 52.5$ g; (c) $m_3 = 49.0$ g

P12.12 (a) $U = 88.2$ N; (b) $D = 58.8$ N

P12.14 (a) $\left[\dfrac{1}{2}m_1 g + \left(\dfrac{x}{L}\right)m_2 g\right]\cot\theta$, $(m_1 + m_2)$g; (b) $\dfrac{\left(m_1/2 + m_2 d/L\right)\cot\theta}{m_1 + m_2}$

P12.16 (a) See ANS. FIG. P12.16; (b) $\dfrac{mg}{2}\cot\theta$; (c) T = $\mu_s mg$; (d) $\mu_s = \dfrac{1}{2}\cot\theta$;
(e) The ladder slips

P12.18 (a) See ANS. FIG. P12.18; (b) 392 N; (c) 339 N to the right; (d) 0;
(e) $V = 0$; (f) 392 N; (g) 339 N to the right; (h) The two solutions agree precisely. They are equally accurate.

P12.20 (a) No time interval. The horse's feet lose contact with the drawbridge as soon as it begins to move; (b) 1.73 rad/s; (c) 2.22 rad/s; (d) 6.62 kN. The force at the hinge is $\left(4.72\hat{\mathbf{i}} + 6.62\hat{\mathbf{j}}\right)$kN; (e) 59.1 kJ

P12.22 (a) $\dfrac{mg\sqrt{2Rh - h^2}}{\left(R - h\right)\cos\theta - \sqrt{2Rh - h^2}\sin\theta}$;

(b) $\dfrac{mg\sqrt{2Rh - h^2}\cos\theta}{(R - h)\cos\theta - \sqrt{2Rh - h^2}\sin\theta}$ and $mg\left[1 + \dfrac{\sqrt{2Rh - h^2}\cos\theta}{(R - h)\cos\theta - \sqrt{2Rh - h^2}\sin\theta}\right]$

P12.24 (a) See ANS. FIG. P12.24; (b) 218 N; (c) 72.4 N; (d) 2.41 m; (e) See P12.24(e) for full explanation.

P12.26 ~ 1 cm

P12.28 (a) 73.6 kN; (b) 2.50 mm

P12.30 1.0×10^{11} N/m^2

P12.32 1.65×10^{8} N/m^2

P12.34 8.60×10^{-4} m

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P12.36 9.85×10^{-5}

P12.38 (a) Rigid object in static equilibrium; (b) See ANS. FIG. P12.38; (c) The woman is at $x = 0$ when n_1 is greatest; (d) $n_1 = 0$; (e) 1.42×10^3 N; (f) 5.64 m; (g) same as answer (f)

P12.40 (a) 0.400 mm; (b) 40.0 kN; (c) 2.00 mm; (d) 2.40 mm; (e) 48.0 kN

P12.42 $\theta = 21.2°$; $T = 1.68$ kN; $R = 2.34$ kN

P12.44 (a) See ANS. FIG. P12.44 for the force diagram and see P12.44(a) for a sample problem statement. (b) The upper hinge exerts 410 N to the left and 442 N up. The lower hinge exerts 410 N to the right.

P12.46 $T = 1.46$ kN; $H = 1.33$ kN; $V = 2.58$ kN

P12.48 (a) 2.71 kN; (b) 2.65 kN; (c) You should lift "with your knees" rather than "with your back"; (d) In this situation, you can make the compressional force in your spine about ten times smaller by bending your knees and lifting with your back as straight as possible.

P12.50 The situation is impossible because the new technique would tip the cabinet over.

P12.52 209 N

P12.54 (a) $n_C = 634$ N, $n_A = 1\,000$ N $- n_C = 366$ N; (b) $C_{AB} = 732$ N, $T_{AC} = 634$ N, and $C_{BC} = 897$ N

P12.56 (a) $T = 133$ N; (b) $n_A = 429$ N, $n_B = 257$ N; (c) $R_x = 133$ N, to the right, $R_y = 257$ N, downward

P12.58 (a) 4 500 N; (b) 4.50×10^6 N/m^2; (c) yes

P12.60 (a) $\frac{YA}{L}$; (b) $YA\frac{(\Delta L)^2}{2L}$

P12.62 (a and b) 120 N, 0.300; (c) 103 N applied at 31.0° above the horizontal at the upper left corner.

P12.64 4.90 cm

P12.66 (a) $93.9d + 125$, in N; (b) See P12.66(b) for full derivation; (c) See P12.66(c) for full derivation; (d) As d increases, T grows larger, θ decreases, and R decreases until about $d = 5.4$ m, then it increases. Note as d increases, the d^2 term predominates.

P12.68 (a) See P12.68(a) for the full explanation; (b) 60.0°; (c) unstable

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